

Online Appendix: Same Storm, Different Boats: Generative AI and the Age Gradient Within Firms

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Latest version: <https://www.ai-econlab.com/research/>. Replication code: <https://github.com/Magnus-L/canaries-sweden>.

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The appendix establishes three things. Across occupations the timing of the relative decline in exposed postings depends on the exposure measure, and no dating of it survives a linear trend (Part II). Between the interim period (December 2022 to December 2023) and the later period (January 2024 to June 2025), the ratio of young to older employment fell 3.9 per cent more at highly exposed employers than at less exposed ones. The decline falls on young women, and the estimate uses no worker's current occupation code (Parts III and VI). A design that classifies workers by their own lagged codes produces a spurious decline among the young, which is why the paper does not use it (Part IV).

Part I: Data and measurement

Exposure to artificial intelligence (AI) comes from the Dynamic AI Occupational Exposure (DAIOE) index of [Engberg et al. \(2024\)](#), which builds on [Felten et al. \(2018, 2021\)](#).¹ The index maps measured progress on AI benchmarks onto the worker abilities that the O*NET database records for each occupation. An occupation scores high when AI has become capable of the things that occupation requires. The index says nothing about whether that capability substitutes for the worker or complements them, and we use it only to rank occupations.

DAIOE is estimated by occupation and by year, and separately for subdomains of AI. We use the generative-AI variant throughout, which aggregates the capabilities behind large language models and image generation, and the 2023 cross-section, the most recent available and the first to reflect the research progress behind the current generation of these systems.

Section [III.2](#) re-estimates the age profile with the GPT-exposure score of [Eloundou et al. \(2024\)](#) in place of DAIOE. Eloundou et al. rate each task as directly exposed (E1), when a large language model alone could complete it in at most half the time, or exposed with complementary software (E2), when that saving needs further software built on the model. Throughout we use their β measure as rated by GPT-4, the share of an occupation’s tasks that are E1 plus one half of those that are E2, which is the measure [Brynjolfsson et al. \(2026\)](#) use, and not α (E1 alone), ζ (E1 and E2 in full) or the human-rated variants. The two scores rank occupations alike. Across the 393 four-digit occupations both score, they correlate at 0.87. Occupations holding 83 per cent of employment in the 2024 occupational register fall on the same side of the top-quartile cut on both measures, and 73 per cent of the employment DAIOE places in its top quartile is in the Eloundou score’s top quartile as well.

At employer level, where the design classifies, 92.5 per cent of DAIOE’s top-quartile employers are in the Eloundou top quartile too, and Section [III.2](#) re-estimates the headline design on that classification.

I.1. Top and bottom occupations

Table [A1](#) lists the ten most and the ten least exposed occupations under the DAIOE generative-AI index.

The exposed end is professional work carried out in language, numbers and code: writers, statisticians, economists, software developers. The unexposed end is work done with the hands, on a site or with the body. This is the variation the paper uses, and the reason an employer is classified by the occupations of its own workforce rather than by its industry.

¹The index is publicly available, with scores by occupation and year, as [Engberg et al. \(2026\)](#), <https://doi.org/10.5281/zenodo.21873968>.

Table A1: Most and least genAI-exposed occupations (DAIOE)

SSYK	Occupation	GenAI pctl
<i>Most exposed (top 10)</i>		
2641	Authors and related writers	99.9
2122	Statisticians	99.7
2121	Mathematicians and actuaries	99.4
2415	Economists	99.2
2512	Software and systems developers	98.9
2145	Chemical engineers	98.7
2111	Physicists and astronomers	98.5
2414	Securities traders and fund managers	98.2
2513	Game and digital media developers	98.0
2112	Meteorologists	97.8
<i>Least exposed (bottom 10)</i>		
9120	Vehicle, window and related cleaners	2.2
2653	Dancers and choreographers	2.0
8350	Ships' deck crew and related workers	1.8
7113	Concrete workers	1.5
8341	Agricultural and forestry machinery operators	1.3
9310	Construction labourers	1.1
8342	Earth-moving machinery operators	0.8
8111	Miners and quarry workers	0.6
7121	Roofers	0.3
3421	Professional athletes	0.1

I.2. Summary statistics

Table A2 describes the employment panel, employer by age band by month from January 2021 to June 2025, with exposure fixed at the employer in 2019. Panel A gives its size at each stage of construction and Panel B the employers behind each comparison the paper draws. Table A3 describes the posting sample on the window the posting estimates use, January 2020 to June 2026.

The panel is balanced and zero-filled. An employer whose young headcount falls to zero still contributes those months, so the extensive margin is inside the coefficient, which is why the estimator is Poisson and not a regression in logs. An employer enters if it holds the young band and at least one older band in some month of the window. One month with one worker in each band meets the condition, and a qualifying floor twelve times higher, five workers for a full year, leaves the estimate almost unchanged (Section III.3). A within-employer age comparison is identified only by employers that hold both ages being compared, so each comparison has a panel of its own. The headline panels are anchored on the young band and the profile panels on the band aged 41–49, and the employer counts in Panel B describe each panel and are not comparable across exhibits.

Table A2: The estimation sample: employer $\times$ age band $\times$ month, 2021:01–2025:06.

<i>Panel A. Cells at each stage of construction</i>					
Band	Outcome	Balanced cells	Zero throughout	Retained	Estimated
22–25	stock	43,763,490	7,216,236 (16%)	36,547,254	24,884,226
22–25	hires	41,088,060	10,351,692 (25%)	30,736,368	20,182,122
22–25	separations	41,088,060	10,633,464 (26%)	30,454,596	20,839,734
26–30	stock	53,649,270	9,443,358 (18%)	44,205,912	28,077,246
26–30	hires	49,355,460	13,040,028 (26%)	36,315,432	—
26–30	separations	49,355,460	13,165,416 (27%)	36,190,044	—
<i>Panel B. Which employers identify which comparison</i>					
Comparison		Condition for entry		Employers	
Scored employers		Incumbents aged 31–69 in 2019		262,089	
Headline, 22–25		Employs 22–25 and an older band		104,217	
Headline, 26–30		Employs 26–30 and an older band		117,090	
Age profile, six bands		Employs 41–49 and another band		153,845	
Age profile, seven bands (65 split)		Employs 22–25 and another band		104,333	
Age profile, eight bands (Figure 2)		Employs 22–25 and another band		104,333	
By field of education, three bands		Employs 41–49, 22–25 or 26–30		110,653	

The panel is balanced over employers, age bands and months and zero-filled. “Zero throughout” counts employer-band cells with no employment in any month; the employer-by-age effect predicts them perfectly, so they are dropped before estimation at no cost to any coefficient. “Retained” is what remains, and “Estimated” the cells at employers with a 2019 exposure score. Flows are estimated at 22–25 only. Panel B gives the employers in each comparison’s own panel. The headline compares each young band with the older bands pooled; the age profiles compare every band with 41–49, the seven-band panel splitting the band aged 50 and over at 65.

Table A3: The posting sample: occupation $\times$ month, 2020:01–2026:06.

	Q1	Q2	Q3	Q4	All
Occupations (SSYK4)	103	99	76	91	369
Occupation $\times$ month cells	7,630	7,612	5,817	7,025	28,084
Ads per month (mean)	99.9	198.1	198.5	181.0	167.2
Ads per month (SD)	212.2	559.8	515.5	364.2	432.6
Ads per month (median)	22	54	49	64	44
Vacancies per month (mean)	255.2	427.7	305.2	256.8	312.7
Vacancies per month (SD)	676.1	1,222.7	796.6	701.8	888.4
DAIOE GAI percentile (mean)	12.4	37.4	61.9	88.2	48.4

Q1 is the least and Q4 the most exposed quartile of the DAIOE generative-AI percentile ranking. Statistics are over occupation-by-month cells with at least one advertisement, the sample of Sections II.2 and II.3.

Part II: Posting and macro analysis

Across occupations, the timing of the relative decline in exposed postings depends on the exposure measure. On DAIOE the decline coincides with the rate rise, and on the rating of Eloundou et al. (2024) it coincides with the launch. Neither date survives a linear trend interacted with exposure or one-digit occupation group $\times$ month effects (Section II.2). Within employers, exposed advertisements decline after the launch whichever measure of

remote work enters (Section II.6; Part V).

II.1. The posting series, its alternatives and the policy cycle

The divergence between equity prices and job postings that opens the paper is not particular to Sweden or to the OMXS30. The S&P 500 rose while the Indeed Hiring Lab index of US postings fell; the OMX Stockholm All-Share Index, which covers every company listed on Nasdaq Stockholm, runs with the OMXS30 throughout; every exposure quartile follows the same path; and the posting decline starts earlier and goes deeper in Sweden, which tightened more sharply. The exhibits behind these statements are in the replication package.

The series rebuilt independently. The posting series was rebuilt independently, with different code and a download taken at a different date. The reconstruction gives $R^2 = 0.998$ against the original occupation-by-month series, and every coefficient holds on both; the estimates on the window to December 2025 were also reproduced in Stata with `reghdfe`, to three decimal places in every specification (Section VI.1).

II.2. Posting estimates by observation window

Table A4 reports the posting difference-in-differences on the window to December 2025 and on the window to June 2026, by OLS and by Poisson. Both windows give a negative rate-rise interaction and an imprecisely estimated post-launch interaction, under either estimator.

Every figure and regression uses the historical archive, which publishes whole quarters once they close. No month is taken from the live feed, whose snapshot endpoint returns only the advertisements live at extraction and so under-counts recent months as advertisements expire.

Table A4: Posting difference-in-differences: the window to December 2025 and the window to June 2026

Window	Estimator	Post-rate-rise $\times$ High	Post-launch $\times$ High	N
January 2020 to December 2025	OLS, ln(ads)	-0.127*** (0.038)	-0.044 (0.037)	25,918
January 2020 to December 2025	Poisson PML	-0.129** (0.051)	-0.060 (0.038)	25,918
January 2020 to June 2026	OLS, ln(ads)	-0.127*** (0.039)	-0.059 (0.038)	28,084
January 2020 to June 2026	Poisson PML	-0.129** (0.051)	-0.060 (0.041)	28,084

Notes: Standard errors clustered by occupation in parentheses. ***, **, * denote significance at 1, 5 and 10 per cent. Occupation and month fixed effects throughout. The Poisson rows are estimated on the same positive cells as the OLS rows; on the balanced panel of 28,782 cells, zero cells included, both extended-window Poisson coefficients and their standard errors move by less than 0.001.

Table A5 varies the construction on the current window. The rate-rise interaction is negative and significant in all nine rows: with positions advertised rather than advertisements as the outcome, without the first half of 2020, without ICT occupations, on a balanced panel, by Poisson, with group-by-month-of-year effects and with the unscored occupations added back. The post-launch interaction is significant only when the high-exposure group is the top tercile rather than the top quartile.

Table A5: Posting robustness, January 2020 to June 2026

	Post-RB $\times$ High	Post-GPT $\times$ High	Occupations	Observations
(1) Baseline	−0.127*** (0.039)	−0.059 (0.038)	369	28,084
(2) Vacancies (positions advertised)	−0.104** (0.044)	−0.052 (0.044)	369	28,084
(3) Excluding January to June 2020	−0.116*** (0.035)	−0.059 (0.038)	369	25,951
(4) Top tercile	−0.093** (0.036)	−0.077** (0.036)	369	28,084
(5) Excluding ICT occupations (SSYK 25)	−0.128*** (0.040)	−0.064 (0.040)	362	27,538
(6) Balanced panel	−0.113*** (0.037)	−0.051 (0.036)	308	24,024
(7) Poisson on counts	−0.129** (0.051)	−0.060 (0.041)	369	28,084
(8) Group by month-of-year effects	−0.130*** (0.039)	−0.062 (0.038)	369	28,084
(9) Unscored codes scored from their group	−0.121*** (0.036)	−0.057 (0.036)	397	30,161

Notes: Equation (1): ln postings on occupation and month fixed effects, High is the top DAIOE quartile, cells with at least one advertisement. Row (2) replaces the outcome with ln of the positions advertised (the sum of each advertisement’s stated number of vacancies); it does not reweight. Row (4) sets High to the top tercile. Row (6) keeps occupations advertised in all 78 months. Row (7) is Poisson pseudo-maximum likelihood on the counts; row (8) adds SSYK 1-digit group $\times$ month-of-year fixed effects. Row (9) adds the 28 non-military occupations the index does not score, each scored as the mean of the scored occupations in its three-digit group; 25 are manager groups that advertisements record without the split into levels. Standard errors clustered by occupation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The exposure ranking and the timing of the posting decline. Table A6 re-estimates Equation (1) with the GPT-exposure score of Eloundou et al. (2024) in place of DAIOE. On the 341 occupations both measures score, the relative decline falls with the rate rise on DAIOE (−0.135, SE 0.042) and after the launch on the Eloundou score (−0.094, SE 0.043), so the date with which the decline coincides depends on the ranking, not the sample. On either measure, neither date survives a linear trend interacted with High or one-digit group $\times$ year-month effects. The posting margin therefore supports a relative decline in exposed occupations after 2022, not a dating of it to one event.

Table A6: Postings with DAIOE and Eloundou et al. (2024) exposure, January 2020 to June 2026

	DAIOE, all occupations			DAIOE, common sample			Eloundou β , common sample		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post-RB $\times$ High	−0.127*** (0.039)	−0.081 (0.051)	−0.040 (0.048)	−0.135*** (0.042)	−0.091 (0.056)	−0.046 (0.054)	−0.023 (0.043)	0.019 (0.058)	0.037 (0.051)
Post-GPT $\times$ High	−0.059 (0.038)	0.008 (0.040)	−0.044 (0.048)	−0.050 (0.042)	0.015 (0.044)	−0.024 (0.052)	−0.094** (0.043)	−0.034 (0.050)	−0.073 (0.049)
Linear trend $\times$ High	Yes			Yes			Yes		
Group $\times$ month FE									
Occupations	369			341			341		
Observations	28,084			25,955			25,955		

Notes: Equation (1) on ln postings, occupation-by-month cells with at least one advertisement. High is the top quartile of the respective measure across all occupations it scores; it is not re-cut on the common sample. The common sample keeps the 341 occupations observed in the posting data and scored by both measures. Columns (2), (5) and (8) add a linear time trend interacted with High; columns (3), (6) and (9) replace month fixed effects with SSYK 1-digit group $\times$ month fixed effects. Standard errors clustered by occupation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

II.3. The posting margin: event study and sensitivity to violations of parallel trends

Figure A1 reports two diagnostics of the posting design on the sample of January 2020 to June 2026: the monthly event study and its pre-period, and the sensitivity of the post-launch average to departures from parallel trends.

Panel (a) plots monthly difference-in-differences coefficients from an event study interacting month dummies with the high-exposure indicator, February 2020 omitted as the reference; 26 coefficients precede the Riksbank’s first rate rise in April 2022. They fluctuate around zero with no systematic trend, but a joint Wald test rejects the null that all 26 are zero ($\chi^2_{26} = 107.2$, $p < 0.01$), and at quarterly frequency, with 8 pre-period coefficients, it still rejects ($\chi^2_8 = 53.0$, $p < 0.01$). The exposure groups responded differently to the macroeconomic cycle before the rate rise, and not only month to month. A rejected pre-test calls for sensitivity analysis rather than abandoning the design (Rambachan and Roth, 2023), just as a passed one would not by itself validate it (Roth, 2022). Under one-digit occupation group $\times$ year-month effects, the post-ChatGPT coefficient is -0.044 (SE 0.048) (Section II.4).

Panel (b) is the relative magnitudes sensitivity analysis of Rambachan and Roth (2023), computed with their HonestDiD procedure on the full vector of monthly coefficients and its clustered covariance. The average post-ChatGPT event-study coefficient for high-exposure occupations over the 43 months from December 2022 to June 2026 is $\hat{\theta} = -0.184$ (SE = 0.057) against February 2020. Re-based to November 2022, the month before the post-period, with the 34 months before it as the pre-period, it is -0.185 (SE 0.051), with a 95 per cent confidence interval of $[-0.284, -0.085]$ under exact parallel trends. Its breakdown value is $\bar{M} = 0.015$. Zero enters the robust interval once the month-to-month change in the differential trend after the launch may reach 1.5 per cent of the largest such change before it, which was 0.287 log points. That bound allows a differential drift of about 0.004 log points a month. Averaged over the 43 months, this is a bias of about 0.09, the distance from the upper end of the interval to zero.

The event-study average, -0.184 , and the pooled post-launch interaction on the same

cells, -0.059 , measure different objects. The average is the gap after the launch against February 2020, so it contains the relative decline that began with the rate rise. In Equation (1) of the paper that decline is the separate term $H_i P_t^{\text{RB}}$ ($\hat{\beta}_1 = -0.127$), and -0.059 is the further shift after the launch, the two summing to -0.186 . The average is also the more sensitive to the reference month and to the pre-trend of panel (a).

The posting margin therefore supports a limited claim: a relative decline in exposed occupations, dated to the rate rise on DAIOE and to the launch on the Eloundou rating (Section II.2), and no additional decline after the launch that survives a small departure from parallel trends.

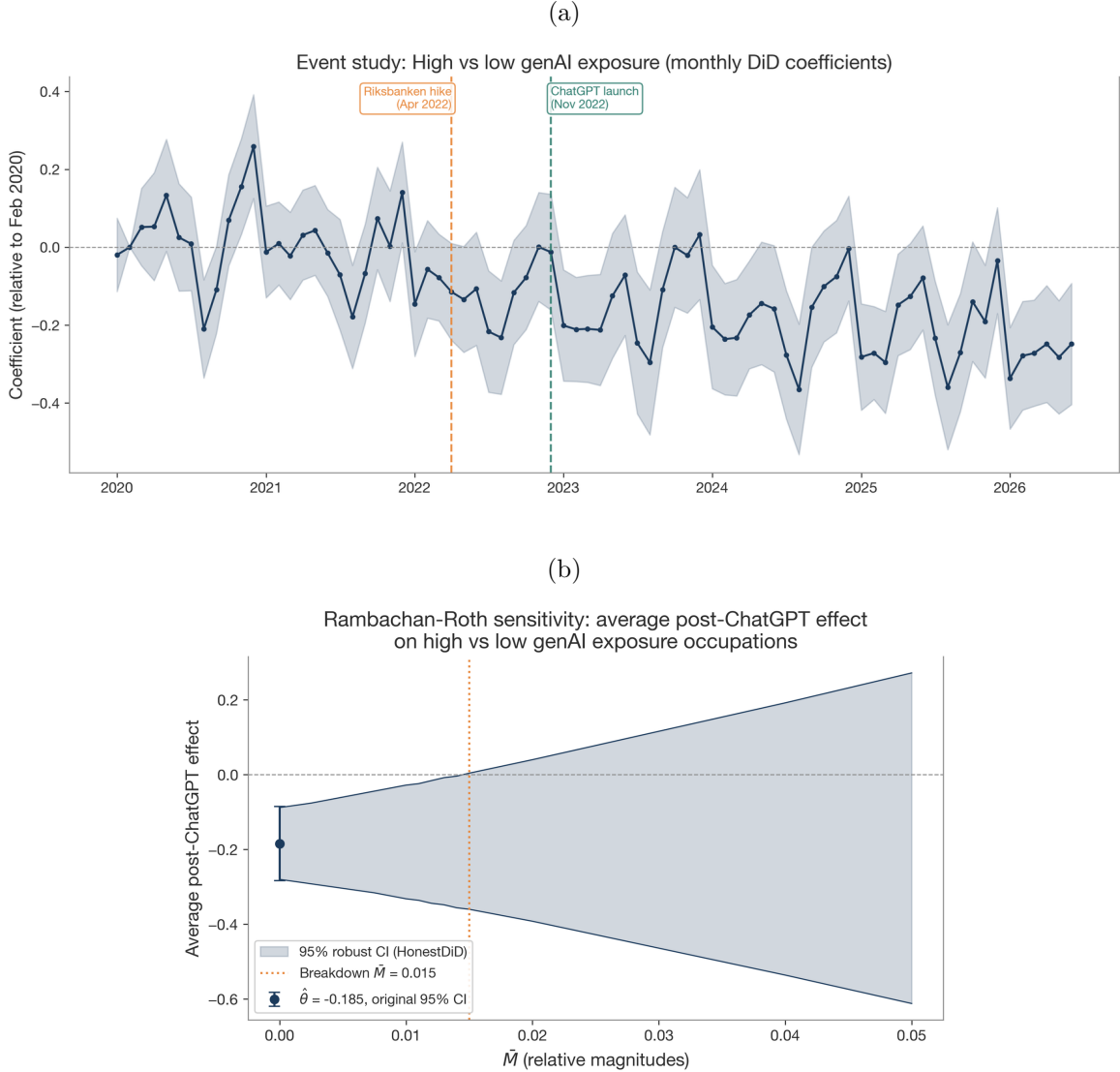


Figure A1: Posting design diagnostics, January 2020 to June 2026. Panel (a): monthly difference-in-differences coefficients, high against low exposure, reference February 2020, with the 95 per cent confidence interval shaded. Panel (b): 95 per cent robust confidence intervals (HonestDiD) for the average post-ChatGPT coefficient under the relative magnitudes restriction, coefficients re-based to November 2022; the point at $\bar{M} = 0$ is the original interval, and the dotted line the breakdown value.

II.4. Seasonality

Swedish hiring is strongly seasonal, and broad occupation groups do not share a seasonal profile, so an event date could pick up the calendar. Table A7 adds one-digit SSYK occupation group $\times$ month-of-year effects to the baseline, absorbing any group-specific seasonal pattern, and neither coefficient moves by more than 0.003. The more demanding one-digit group $\times$ year-month effects absorb most of the identifying variation, and under them neither coefficient is distinguishable from zero. The employment design faces the same hazard in a sharper form and carries calendar-quarter terms in the specification itself (Section III.2).

Table A7: Posting estimates under seasonal controls

Specification	Post-rate-rise $\times$ High	Post-launch $\times$ High
Baseline (occupation and month FE)	−0.127*** (0.039)	−0.059 (0.038)
+ one-digit occupation group $\times$ month-of-year	−0.130*** (0.039)	−0.062 (0.038)
+ one-digit occupation group $\times$ month of sample	−0.040 (0.048)	−0.044 (0.048)
Poisson, with one-digit group $\times$ month-of-year	−0.135** (0.054)	−0.059 (0.042)

Notes: Standard errors clustered by occupation in parentheses. ***, **, * denote significance at 1, 5 and 10 per cent.

II.5. Exposure deciles on the posting margin

Table A8 reports the posting design with a full set of decile $\times$ period interactions, the median decile omitted, to ask whether a finer cut than the top quartile shows a monotone gradient. It does not. Post-ChatGPT coefficients are positive across most of the distribution and peak in the middle deciles, and the only negative decile is the eighth. Each decile holds about 37 occupations and the standard errors are roughly twice those at quartile level, so the occupation-level posting margin cannot support a fine gradient, and the paper reads it at the quartile cut.

Table A8: Posting estimates by exposure decile

Exposure decile	Post-rate-rise $\times$ decile	Post-launch $\times$ decile
1	0.003 (0.082)	0.117 (0.084)
2	0.081 (0.077)	0.093 (0.083)
3	0.171** (0.076)	0.094 (0.078)
4	0.020 (0.071)	0.175** (0.073)
6	0.000 (0.070)	0.164** (0.078)
7	0.001 (0.076)	0.195** (0.082)
8	0.055 (0.075)	−0.018 (0.088)
9	−0.066 (0.078)	0.045 (0.070)
10	−0.123* (0.069)	0.049 (0.072)

Notes: Standard errors clustered by occupation in parentheses. Decile 5, the median, is the omitted reference. ***, **, * denote significance at 1, 5 and 10 per cent.

II.6. Rival explanations: rate sensitivity and remote work

Figure A2 sets two rivals against an AI reading of the posting margin: that exposed occupations are more sensitive to monetary policy, and that exposure stands in for the ability to work remotely. The second is the argument of Lambert and Schindler (2026): generative-AI exposure is strongly correlated with the post-pandemic shift to working from home, remote work raises the cost of supervising and developing juniors, and entered together the remote-work term stands while the AI term attenuates towards zero.

Panel (a) plots a revealed measure of sensitivity to monetary policy, the log change in mean postings between the first quarter of 2022 and May to November 2022, against exposure. The window closes before the launch, so any decline inside it predates ChatGPT. Exposure is not detectably correlated with it ($r = 0.035$, $p = 0.51$ across the 353 occupations with postings in both windows; -0.020 weighted by posting volume). In the United States Brynjolfsson et al. (2026) find a negative correlation between AI exposure and the interest-rate sensitivity measure of Zens et al. (2020).

AI-exposed occupations overlap substantially with occupations that can be done at home (Dingel and Neiman, 2020; Lambert and Schindler, 2026), so the discriminating variation is the off-diagonal: occupations exposed but not teleworkable, and teleworkable but not exposed. Panel (b) reads it directly, splitting the Dingel and Neiman (2020) classification at its median and re-estimating Equation (1) of the paper in each half. In the 187 teleworkable occupations the post-launch coefficient is indistinguishable from zero and the decline sits in the rate-rise term. In the 182 occupations that cannot be done from home, the post-launch coefficient is -0.216 and significant at one per cent.

Lambert and Schindler measure realised rather than feasible remote work: the share of an occupation's postings in 2021 and 2022 that offer remote or hybrid work (Hansen et al., 2023). Table A9 sets two realised measures beside feasibility. The first is the series they use, here its United States postings (they pool four countries), crosswalked to SSYK. The second is read from the text of the Swedish advertisements, with a precision of about 94 per cent and a recall of about 90 per cent on hand-read samples. Both correlate with AI exposure at 0.75 to 0.79 across occupations, as closely as feasibility does. Across occupations the two realised measures disagree. Each score alone predicts a post-launch decline. Entered together on the Swedish measure, the AI coefficient falls from -0.064 (0.017) to -0.016 (0.022) and remote work keeps -0.070 (0.021). On their own measure, the AI coefficient stays at -0.066 (0.023) and remote work falls to 0.003 (0.024). This specification has a single post-launch period, which also contains the fall that began with the rate rise. It therefore asks only which score absorbs the other, and across occupations the data do not settle that question.

Within employers, in the comparison Part V draws on public advertisements, the decline does not depend on the measure. With each employer's advertisements cut by AI exposure, the post-launch decline in exposed advertisements is -0.158 (0.013). Cutting them also by realised remote work leaves it at -0.179 (0.016) on the Swedish measure and -0.170 (0.014) on Lambert and Schindler's. It is -0.129 (0.022) in the 9,324 employers that did not advertise remote work in 2021 and 2022 and -0.163 (0.014) in those that did, a difference that is not significant. Not advertising remote work is a diagnostic rather than a measure, since such an employer may still have used it, but a mechanism running through remote firms would be hard to reconcile with a decline of that size in the first group. Across occupations,

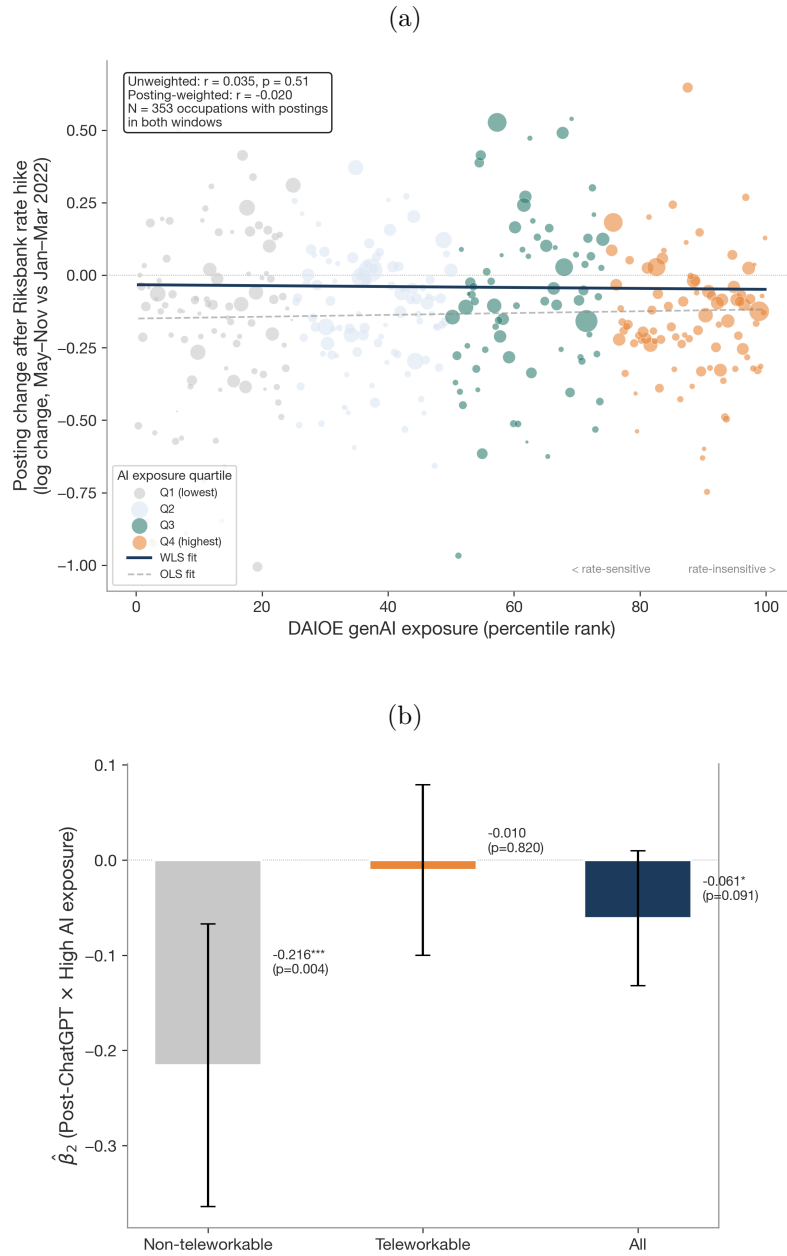


Figure A2: Two rival explanations for the posting pattern. Panel (a): revealed monetary policy sensitivity, the log change in postings from January to March 2022 to May to November 2022, against generative-AI exposure, for the 353 occupations with postings in both windows; bubble size is posting volume, and the printed p -value takes the occupation as the unit. Panel (b): the post-ChatGPT coefficient estimated separately by teleworkability (Dingel and Neiman, 2020), January 2020 to June 2026, with 95 per cent confidence intervals, on the log-plus-one outcome of the split fits; for all occupations it gives -0.061 , against -0.059 on $\ln A$ in Table A4.

Table A9: AI exposure and three measures of remote work on the posting margin

	Dingel and Neiman (feasible) (1)	Platsbanken (realised, Sweden) (2)	Hansen et al. (realised, US) (3)
Correlation with AI exposure	0.79	0.75	0.79
<i>A. Across occupations, all advertisements (OLS)</i>			
Below the median of the measure	−0.216*** (0.076)	—	—
Above the median of the measure	−0.010 (0.046)	—	—
AI exposure alone	—	−0.064*** (0.017)	−0.064*** (0.017)
Remote work alone	—	−0.080*** (0.016)	−0.047*** (0.017)
Joint: AI exposure	—	−0.016 (0.022)	−0.066*** (0.023)
Joint: remote work	—	−0.070*** (0.021)	0.003 (0.024)
Occupation-months	28,084	28,084	28,084
<i>B. Within employer, all advertisements (Poisson)</i>			
AI exposure alone	—	−0.158*** (0.013)	−0.158*** (0.013)
Joint: AI exposure	—	−0.179*** (0.016)	−0.170*** (0.014)
Joint: remote work	—	0.039* (0.022)	0.027 (0.019)
AI exposure, employers advertising remote	—	−0.163*** (0.014)	—
AI exposure, employers not advertising remote	—	−0.129*** (0.022)	—
Difference	—	−0.033 (0.026)	—
Employers	—	12,141	12,141
<i>C. Across occupations, entry-level advertisements (OLS)</i>			
Joint: AI exposure	—	−0.120*** (0.038)	−0.168*** (0.042)
Joint: remote work	—	−0.037 (0.028)	0.018 (0.029)
Occupation-months	—	12,159	12,159

Post-launch coefficients (from December 2022); standard errors, in parentheses, clustered by occupation (panels A and C) or employer (panel B). Column (1): the [Dingel and Neiman \(2020\)](#) teleworkability classification, a measure of feasibility, split at its median as in Figure A2(b), with Equation (1) estimated in each half. Column (2): the share of an occupation’s advertisements in 2021 and 2022 that offer remote or hybrid work, read from the advertisement text. Column (3): the same share in United States postings ([Hansen et al., 2023](#)), the source [Lambert and Schindler \(2026\)](#) use, which pools four countries, crosswalked to SSYK. Correlations are across the 369 occupations, weighted by 2024 employment. Panels A and C follow Lambert and Schindler: one post-launch dummy interacted with each standardised score, alone and together, with occupation and month effects. Panel B is the design of Section V.1, with each employer’s cells cut by the top quartile of AI exposure and of remote work. Its employer split, possible only in column (2), separates employers by whether any of their own 2021 to 2022 advertisements offered remote work. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

among entry-level advertisements, the margin closest to junior hiring, exposure keeps -0.120 (0.038) and -0.168 (0.042) on the two measures, and remote work is indistinguishable from zero on both.

The employment margin offers no such comparison: averaged over an employer’s incumbents, AI exposure and teleworkability correlate at 0.88, and only 14 per cent of employers lie off the diagonal of a median cut on both. Section III.2 enters both in the headline design.

II.7. Advertisement sample accounting

Table A10 reports, year by year to June 2026, how many advertisements were published and how many were removed for a missing occupation field, an invalid occupation code, a date outside the window, or duplication on the advertisement identifier. It separates a fall in what employers advertise from a fall in what the collection keeps.

About one advertisement in a hundred is lost. Every missing occupation field falls in 2020 or 2021. From 2022 to June 2026 nothing is lost to coding, and no advertisement is removed for an invalid code, so the restriction binds only through missing fields and duplicates, both recorded at publication. The first half of 2026 comes from the two closed-quarter files and is processed with the same classification. Of its advertisements dated outside January to June 2026, 2,657 were published in 2025, all but 100 of them already in the 2025 file, and 187 fall in July 2026. None repeats an identifier kept in 2020 to 2025.

From 400 occupations to 369. The larger restriction operates on occupations. Advertisements published between January 2020 and June 2026 carry 400 distinct four-digit SSYK codes, and the regression sample holds 369. The missing 31 are the codes the DAIOE index does not score once it is crosswalked from O*NET to SSYK: 3 are the military occupations of major group 0, which O*NET does not cover, 26 are managerial occupations in major group 1, 25 of which advertisements record without the split into a first and a second level that the index scores separately, and 2 are single codes in major groups 3 and 7. No occupation is lost for having no advertisements. Those 31 codes carry 3.7 per cent of advertisements. Scoring each of the 28 non-military codes as the mean of the scored codes in its three-digit group adds them back and leaves both interactions close to the baseline (Table A5, row 9).

Table A10: Advertisement sample accounting, January 2020 to June 2026

Year	Raw ads	No occ. field	Invalid code	No date	Out of range	Duplicate	Parse error	Kept
2020	490,713	106	0	0	2	0	0	490,605
2021	730,824	4,845	0	0	0	2,407	0	723,572
2022	1,082,795	0	0	0	0	26,691	0	1,056,104
2023	1,028,074	0	0	0	0	4,646	0	1,023,428
2024	717,094	0	0	0	0	4,018	0	713,076
2025	582,241	0	0	0	0	2,853	0	579,388
2026 (Jan–Jun)	294,095	0	0	0	2,844	1,650	0	289,601
Total 2020–2025	4,631,741	4,951	0	0	2	40,615	0	4,586,173
Total to June 2026	4,925,836	4,951	0	0	2,846	42,265	0	4,875,774

Notes: Advertisements published on Platsbanken, from the JobTech historical bulk files for 2020 to 2025 and the two closed-quarter files for 2026, which cover January to June only. Categories are applied in the order shown and are mutually exclusive, except that for 2026 duplicates are removed before the window is applied, as in the estimation sample. Duplicates are exact repeats of an advertisement identifier. Out of range for 2026 counts advertisements dated outside January to June 2026.

II.8. Occupation-code coverage over time and by source

Table A11 reports the share of advertisements carrying a valid SSYK code, on which the exposure quartile depends, by year and by the channel through which the advertisement reached Platsbanken. Coverage is complete in every channel in every year from 2022 to the first half of 2026. It dips only in 2021, to 99.3 per cent, and the lowest months of the window, June and July 2021, still code more than nineteen advertisements in twenty. From 2022 there is no uncoded mass left for the exposure groups to differ over, so coverage cannot differ between them in any month after the rate rise.

Figure A3 draws the three monthly series. The number of occupations with at least one advertisement in a month stays between 374 and 396 from January 2020 to June 2026, with no trend and no break at either event date, and occupation-month cells with no advertisement average one per cent in the most exposed quartile, in no month of 2026 exceeding two of its 91 occupations.

Table A11: Share of advertisements with a valid SSYK code, by year and source

Year	AIS	Annonsera	JobPosting	No source	Platsbanken DXA	All sources
2020	100.0	100.0	–	99.9	100.0	100.0
2021	99.0	99.4	–	–	99.3	99.3
2022	100.0	100.0	–	–	100.0	100.0
2023	100.0	100.0	–	–	100.0	100.0
2024	100.0	100.0	–	–	100.0	100.0
2025	100.0	100.0	100.0	–	100.0	100.0
2026 (Jan–Jun)	100.0	100.0	100.0	–	100.0	100.0

Notes: Per cent. 2026 covers January to June. Duplicates are included, since coverage is a property of the data. A dash denotes a channel with fewer than 100 advertisements in that year. “No source” are advertisements whose delivery channel is not recorded, almost all of them in 2020.

Part III: Employment analysis

On the baseline index, the young-to-older employment ratio at ages 22 to 25 falls 3.9 per cent more at highly exposed employers between the interim and later periods, and the shortfall is concentrated among young women. This part reports the raw counts, the design from its headline and timing to its splits by age, sex and field of education, the rival explanations it can test, the precision of the employer score, and a comparison with related studies.

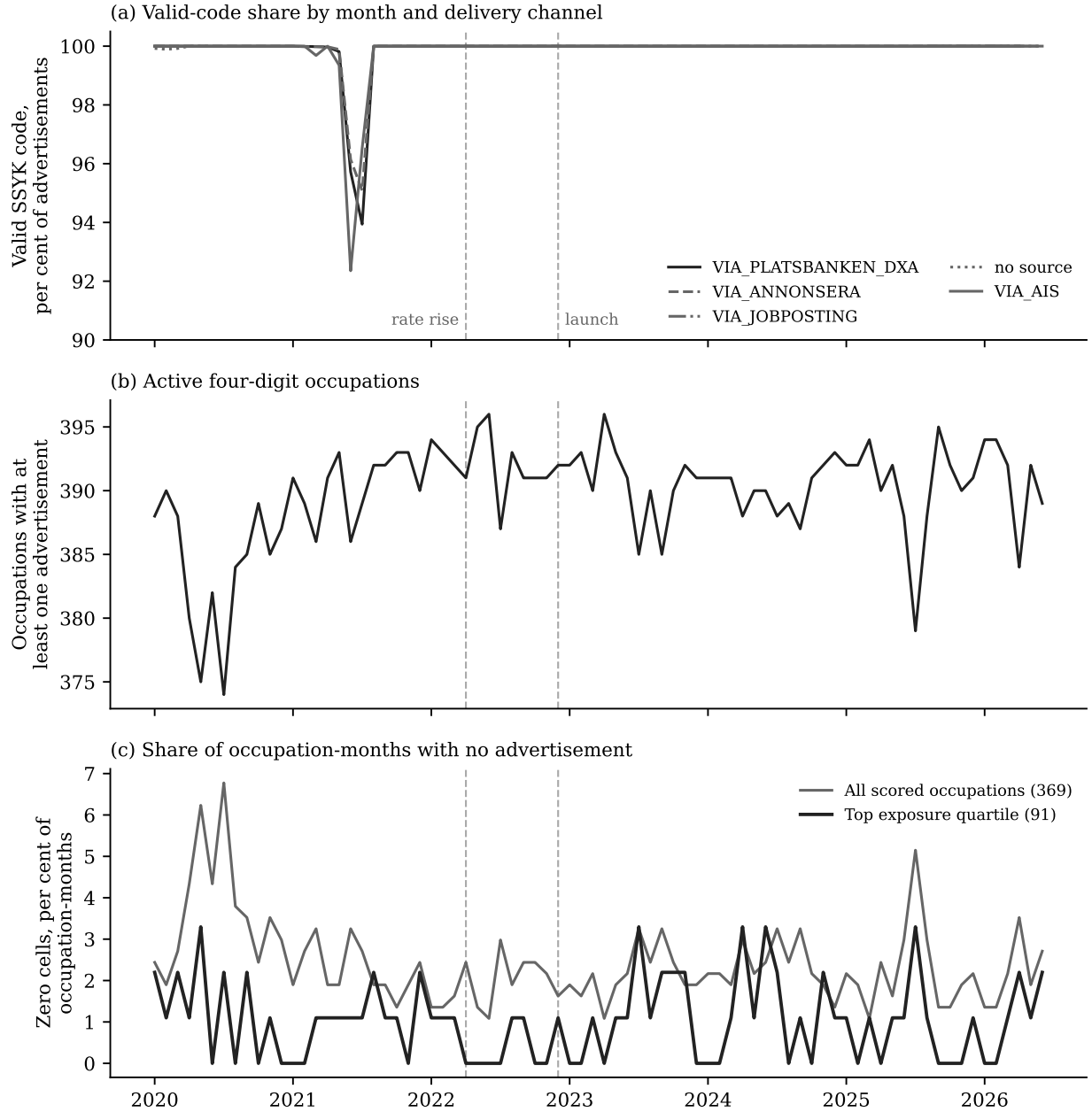


Figure A3: Posting coverage by month, January 2020 to June 2026. (a) Share of advertisements with a valid SSYK 2012 code, by delivery channel. (b) Four-digit occupations with at least one advertisement in the month. (c) Share of occupation-months with no advertisement on the balanced grid of DAIOE-scored occupations, all and the top exposure quartile. Dashed lines: the first rate rise (April 2022) and the first full month after the launch (December 2022). Duplicates included.

III.1. Employment by age group and AI exposure

Table A12 gives the raw counts behind the design of Section III.2: the change from the two years before the later period, 2022 and 2023, to the later period itself, by exposure quartile and age band, with no controls. Panel A counts total headcount per month and

Panel B the mean headcount per employer-month in which the band is present, and the set of employers is not held fixed in either. Swedish university completion is late, with a median graduation age of 27 to 28, so the band aged 22–25 captures very-early-career work and the band aged 26–30 is where many graduate careers begin.

Employment of the young fell at exposed and less exposed employers alike: at both young bands the fall in the most exposed quartile is close to the fall in the least exposed. What separates exposed employers is the other end of the age distribution. The two most exposed quartiles grew their workforce aged 50 and over, by far the most in the top quartile, while the two least exposed shrank theirs. The per-cell means of Panel B rise in every band and quartile because they condition on the cell being populated.

Hiring fell in every quartile at every age below 50 (Panel C). At 22–25 it fell by about as much as at 41–49, and at the most exposed employers by no more than at the least exposed. Before 2024 the young were hired at two to four times the monthly rate of those aged 41–49 (Panel D), so an equal proportional fall in hiring reduces their headcount more.

The paper’s estimates are therefore statements about relative age structure, not counts of jobs: a negative coefficient means that the employment of the young fell relative to that of their own older colleagues inside the same firm.

The later period. The later period runs from January 2024 to June 2025. Its start is set by the spread of AI use in the Swedish economy, not by adoption measured at any one employer. ChatGPT launched at the end of 2022, but availability is not use. Among firms with ten or more employees, the share using at least one AI technique rose from 10 per cent in Statistics Sweden’s January 2023 reading to 25 per cent in January 2024 ([Statistics Sweden, 2026c](#)), and by May 2024 a majority of professional employees reported using AI in their work ([Akavia, 2026](#)). The window opens in January 2024, the first reading at which use is common. The date is when the survey measured the rise, not a break in the data, and [Section III.2](#) shows that the decline is gradual.

Table A12: Descriptive counterpart of the headline design: employment and hiring by exposure quartile and age band, January 2022 to December 2023 against the later period (January 2024 to June 2025).

Exposure quartile of the employer (2019 occupation mix)				
<i>Panel A. Total monthly headcount, per cent change</i>				
Age band	Q1 (least)	Q2	Q3	Q4 (most)
22–25	–8.1	–5.9	–2.8	–8.5
26–30	–13.0	–10.0	–7.5	–11.5
31–34	–8.6	–3.8	–1.5	–3.8
35–40	–2.5	+3.1	+4.1	+2.0
41–49	–6.0	–2.8	–1.1	–4.6
50+	–0.8	–0.4	+1.5	+5.9
<i>Panel B. Mean headcount per populated employer-month, per cent change</i>				
Age band	Q1 (least)	Q2	Q3	Q4 (most)
22–25	+2.1	+2.6	+4.9	+2.5
26–30	+2.1	+2.9	+4.4	+1.0
31–34	+5.2	+6.9	+7.8	+5.8
35–40	+8.8	+13.2	+12.4	+14.4
41–49	+5.5	+7.4	+10.3	+8.8
50+	+6.5	+5.7	+7.7	+11.6
<i>Panel C. Total monthly hires, per cent change</i>				
Age band	Q1 (least)	Q2	Q3	Q4 (most)
22–25	–12.0	–8.2	–4.8	–10.4
26–30	–18.1	–15.1	–12.5	–20.5
31–34	–15.4	–12.5	–10.0	–17.0
35–40	–11.3	–5.0	–3.6	–11.4
41–49	–12.4	–8.9	–6.8	–14.0
50+	–7.3	–3.8	–2.1	+1.1
<i>Panel D. Monthly hire rate, January 2022 to December 2023, per cent</i>				
Age band	Q1 (least)	Q2	Q3	Q4 (most)
22–25	10.9	11.5	13.8	15.1
26–30	8.0	7.9	8.9	7.0
31–34	6.5	6.3	6.9	5.3
35–40	5.8	5.0	5.6	4.7
41–49	5.0	3.9	4.4	4.1
50+	4.6	3.7	3.9	5.0

Per cent change in average monthly headcount or hires from January 2022 to December 2023 to January 2024 to June 2025, by the employer’s exposure quartile; for example, –8.5 means that the monthly headcount aged 22–25 summed over top-quartile employers was 8.5 per cent lower in the later period. Panel A: total headcount per month over every employer in the quartile, zeros included. Panel B: mean headcount per employer-month with at least one worker in the band, which conditions on a populated cell. Panel C: total hires per month, a hire being a worker present at an employer in a month and absent from it the month before. Panel D: monthly hires as a percentage of monthly headcount over January 2022 to December 2023. No controls enter.

III.2. The design the paper reports

The estimates in Section 3 of the paper compare young workers with their older colleagues inside the same employer. An employer is classified once, by the employment-weighted mean generative-AI percentile of the occupations its incumbents aged 31 to 69 held in 2019, and no worker under 31 is ever classified. The exposure, the outcome and the sample read no occupation register file after 2019, so truncating the register at any later year would leave every input unchanged. A design that classifies each worker by their own latest code, by contrast, inherits codes stale enough to produce a spurious decline among the young, which Part IV measures. Later individual registers supply only birth year and sex, so workers first registered in Sweden in 2024 or 2025 are never counted (Part VI). Section IV.4 reports which codes the 2019 score reads.

What the within-employer contrast identifies. The workers whose occupations define an employer’s exposure are also, in later months, part of the older group the young are compared with. The score is fixed in 2019 and the outcome runs from January 2021, so no cell enters both. What the construction cannot exclude is selection: an employer whose older workers held exposed occupations in 2019 may differ later in ways that move its age structure for reasons other than AI. The employer-by-month effects and the calendar terms absorb what moves an employer as a whole, and the shocks that could act on one age inside it are tested below.

Differencing two ages inside one employer also has an economic rationale. [Chandar and Klein Teeselink \(2026\)](#) show that the employment gap between an adopting firm and an otherwise identical non-adopter, for a given type of worker, is the sum of a shifter specific to that type and a scale term common to every type in the firm. The scale term is positive wherever adoption lowers unit cost, so a type of worker can lose tasks to AI and still grow at an adopting firm. Differencing two types inside the same firm removes the scale term, so a coefficient here describes the young relative to their older colleagues and not the level of employment at either age.

Whether the measure picks out employers that use AI. Figure A4 sets the score, which never reads a survey, against two independent sources. In Statistics Sweden’s survey of firms, top-quartile employers are 20.9 percentage points more likely to report using AI in 2023, and in its 2024 survey of workers their employees are 22.5 points more likely to report using generative AI. The score also predicts AI use that predates ChatGPT: the differential is 18.2 points in 2019 and not distinguishable from 2023. Exposed employers were already the more AI-intensive before generative AI existed, so a comparison of exposed with less exposed employers would compare firms that already differed. Every employment estimate is therefore drawn inside the employer, and exposure is nowhere treated as exogenous.

The periods, and the headline they define. The headline compares the later period, January 2024 to June 2025, with the interim period, December 2022 to December 2023, which is its comparison period. The interim period serves as the comparison because it follows the launch but precedes widespread use: the share of firms reporting AI use rose from 10 to 25 per cent between January 2023 and January 2024. It does not follow the tightening, which continued until September 2023; the design only keeps the tightening months before

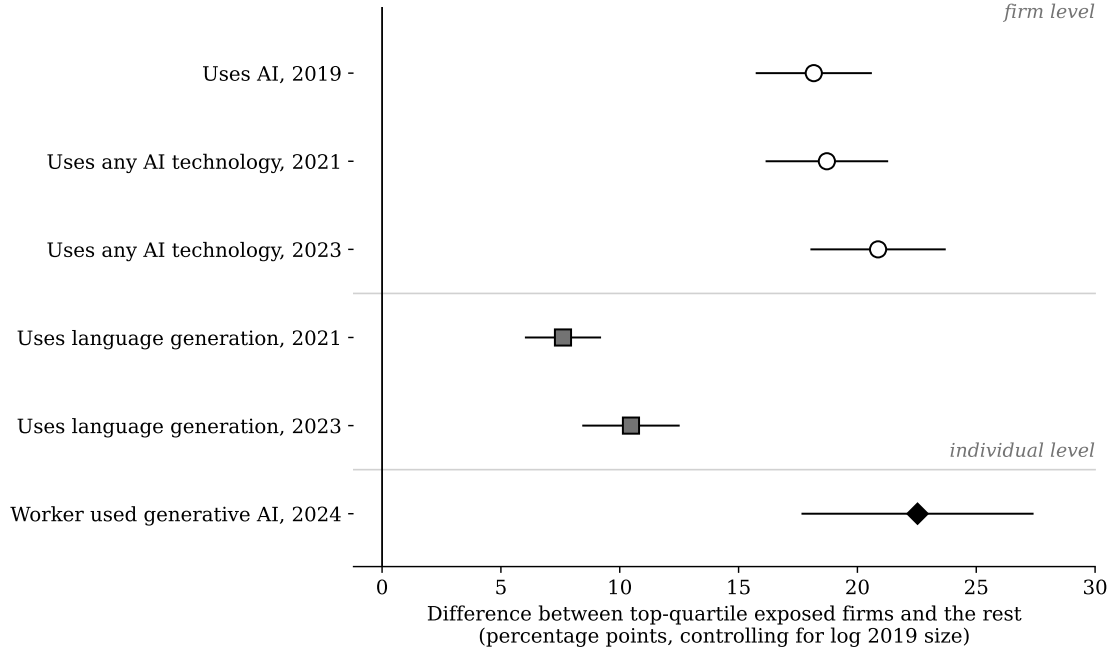


Figure A4: Top-quartile employers' reported AI use: Statistics Sweden's ICT survey of firms (any AI use 2019, 2021 and 2023; language generation 2021 and 2023) and its 2024 worker survey. Points are the top-quartile differential in percentage points with 95 per cent intervals; firm rows control for log 2019 employment and the worker row is survey-weighted. The waves are separate cross-sections, and the 2019-to-2023 difference has a standard error of 1.9 points. The 2023 firm row rests on 3,587 firms and the worker row on 2,285 respondents.

the launch out of both periods. The headline, $\tau = b_L - b_I = -0.040$ or 3.9 per cent, is the change in the young-to-older ratio inside the same employer from the interim to the later period, at highly exposed employers relative to less exposed ones.

Equation (2) divides the panel into four periods, and Table A13 sets out what each coefficient measures in two equivalent parameterisations. The interim period covers the thirteen months after the launch of ChatGPT and before the later period. The reference period, January 2021 to March 2022, is the fifteen months against which the panel is normalised. The tightening months, April to November 2022, have their own indicator, so that the months of tightening before the launch enter neither the interim nor the later period. The paper writes Equation (2) with three mutually exclusive period indicators, so that b_R , b_I and b_L are each measured against the reference period. An equivalent parameterisation, the cumulative form, keeps the tightening indicator switched on from April 2022 to the end of the panel, so that γ_0 (interim) and γ_2 (later) are measured against the tightening months. The two describe the same fitted means, with $b_I = \gamma_1 + \gamma_0$, $b_L = \gamma_1 + \gamma_2$ and $\tau = \gamma_2 - \gamma_0$.

The two forms give three readings of the later period. Against the months before the rate rise, the level is $\hat{b}_L = \hat{\gamma}_1 + \hat{\gamma}_2 = -0.042$. Against the tightening months, the step is $\hat{\gamma}_2 = -0.058$, large partly because the young gained during those months. Against the interim period, the period immediately before it, the headline is $\hat{\tau} = \hat{\gamma}_2 - \hat{\gamma}_0 = -0.040$. The headline is therefore about the same as the level against the reference period and smaller than the step from the tightening months.

Table A13: The four periods of Equation (2) and what each coefficient measures.

Period	Months	Paper, against the reference	Cumulative, terms on
Reference	Jan 2021 to Mar 2022	none	none
Tightening	Apr 2022 to Nov 2022	b_R	γ_1
Interim	Dec 2022 to Dec 2023	b_I	γ_1, γ_0
Later	Jan 2024 to Jun 2025	b_L	γ_1, γ_2

The reference runs from the start of the panel to the month before the first rate rise, so that 2019, the scoring year, stays out of the outcome window and the pandemic year out of the reference. The three calendar-quarter terms are on throughout. $\hat{\tau} = \hat{b}_L - \hat{b}_I = \hat{\gamma}_2 - \hat{\gamma}_0$ is the headline. In the cumulative form, γ_0 and γ_2 are measured against the tightening months (Table A14 and Table A15).

The period coefficients. Table A14 reports the coefficients of the paper’s parameterisation, each against January 2021 to March 2022, and τ . Table A15 collects the steps from the tightening months, the drift, the profile, the flows and the sexes. In the later period the ratio sits below its level before the rate rise at both bands. At 22–25 it rose during the tightening months, was back near its level before the rate rise on average over the thirteen months after the launch, and ends 0.042 below that level.

Table A14: The period coefficients of Equation (2), each period against January 2021 to March 2022, and the headline τ .

Term	22–25	26–30
Tightening months, April to November 2022 (b_R)	+0.0159* (0.0075)	+0.0184* (0.0044)
Interim period (b_I)	−0.0020 (0.0135)	+0.0104 (0.0080)
Later period (b_L)	−0.0419* (0.0189)	−0.0298* (0.0123)
Later minus interim (τ)	−0.0399* (0.0102)	−0.0403* (0.0067)
Employers	104,217	117,090

Poisson with employer-by-month, employer-by-age and month-by-age effects and the calendar-quarter terms; exposure is the employer’s 2019 occupation mix; standard errors clustered by employer. The three period indicators are mutually exclusive, so each row is a level against January 2021 to March 2022. The cumulative parameterisation of Table A15 describes the same fitted means. The fourth row is the headline of Table 1 of the paper. Each band is fitted on its own panel. * $p < 0.05$.

A gradual decline, and where the boundary falls. The period estimates show the ratio rising through the tightening months; Figure 3 of the paper traces the quarterly path from the launch, turning down after the launch quarter and declining through 2023 and 2024. The January 2024 boundary is set from outside the employment data, by the timing of a survey rather than by a break in the data: Statistics Sweden’s enterprise survey measures AI use in January of each year (Statistics Sweden, 2023, 2024), so the 10 and 25 per cent are readings of January 2023 and January 2024. The rise between them took place during 2023, so the interim period is a year of rising use rather than an untreated baseline.

No month is dropped. Every month after the launch is estimated, and the early movement enters as $\hat{\gamma}_0$ and is subtracted, not set to zero. Moving the boundary to July 2023, October 2023, April 2024 or July 2024, with the interim period always running from the launch, gives τ between −0.041 and −0.039 at 22–25 and between −0.050 and −0.040 at 26–30, largest at

Table A15: Components behind the headline: steps from the tightening months, drift, the age profile, flows and the sexes

	Estimate (employer SE)	Industry SE
<i>Panel A. Step from the tightening months to the later period ($\hat{\gamma}_2$)</i>		
Ages 22–25, against the older bands pooled	−0.0578 (0.0155)	(0.0307)
Ages 26–30, against the older bands pooled	−0.0482 (0.0104)	(0.0208)
<i>Panel B. Pre-launch drift per month, January 2021 to November 2022</i>		
Ages 22–25	+0.0004 (0.0008)	
Ages 26–30	+0.0016 (0.0004)	
<i>Panel C. The profile in the later period, against 41–49 alone</i>		
22–25	−0.0192 (0.0125)	
50 and over	+0.0675 (0.0063)	
<i>Panel D. Flows at 22–25, step from the tightening months</i>		
Hires	−0.0179 (0.0406)	
Separations	+0.0540 (0.0225)	
Hires, later minus interim (τ)	+0.0084 (0.0342)	(0.0498)
Separations, later minus interim (τ)	+0.0300 (0.0144)	(0.0322)
<i>Panel E. The sexes at 22–25, step from the tightening months</i>		
Young men	−0.0171 (0.0179)	(0.0352)
Young women minus young men	−0.0858 (0.0142)	(0.0174)
Young women	−0.1029 (0.0163)	(0.0295)
Young women minus young men, within broad education tracks	−0.0731 (0.0138)	

Poisson with the effects and calendar terms of Equation (2), exposure the employer’s 2019 occupation mix, clustered by employer and, in the second column, by three-digit industry. Panels A, D and E report the cumulative parameterisation, the step from April to November 2022 to January 2024 to June 2025, a different contrast from the headline τ of Table 1 of the paper; the period coefficients behind τ are in Table A14. The drift is a linear monthly trend. Hires and separations are relative counts, not hazards. The within-track row is the female differential estimated within each broad field of the worker’s own education and weighted by young women’s track shares (Table A20).

the two later boundaries. Because both periods move with the boundary, a gradual decline returns a similar estimate wherever the boundary falls. The sweep therefore shows that the estimate does not depend on where the boundary is placed. It does not date the start of the decline.

Table A16 reports the decline on contrasts that need no boundary. A second fit replaces the interim and later-period indicators with one indicator per quarter from the launch onward, keeping the Riksbank and calendar terms, so that every quarter is read against the tightening months net of the calendar cycle. Its first bin is December 2022 alone, which is why Figure 3 draws that point hollow. Calendar 2024 lies −0.034 (SE 0.008) below calendar 2023 at 22–25 and −0.024 (0.006) at 26–30, and the fall is largest in the second halves of the two years. The two quarters either side of the boundary differ by +0.004 (0.009) at 22–25, so no discontinuity is detected at January 2024.

Timing has a limit. Keeping the Riksbank indicator switched on holds the monetary contribution at the level it reached during the tightening months and does not model the dynamics of monetary policy. If tightening reached the young at exposed employers with a lag, the later coefficients would absorb it, which is why the credit channel is tested directly below. The ratio rose during the tightening, however, and the shortfall persists through

Table A16: The decline read off the quarterly path, on contrasts that do not depend on where the later period is judged to begin.

Contrast	Ages 22–25	Ages 26–30
Calendar 2024 against calendar 2023	−0.0342* (0.0084)	−0.0235* (0.0056)
Second half of 2024 against second half of 2023	−0.0449* (0.0093)	−0.0278* (0.0058)
2024Q1 against 2023Q4, the quarters either side of the boundary	+0.0041 (0.0091)	−0.0046 (0.0045)

Linear combinations of the quarterly coefficients of Figure 3 of the paper, with standard errors from that fit’s clustered covariance. The first two rows compare periods of equal length and identical calendar-quarter composition without using the January 2024 boundary; the third compares the quarters either side of that boundary, net of the estimated calendar terms. The first row is the boundary-free counterpart of the headline, weighting quarters equally where the pooled coefficient weights employer-months. * $p < 0.05$.

more than a year of easing from May 2024.

The pre-period. Figure A5 and Table A17 report the quarterly path with no calendar terms from 2019Q1, and the lower panels of Table A17 the drift tests and the backdated placebo. The young-to-older ratio has a strong calendar cycle, highest in the fourth quarter, so the path is read within quarters across years and tested by a linear drift fitted with the calendar terms and the tightening window. The specification behind Figure 3 of the paper cannot be extended into the pre-period. If every quarter before the launch had its own indicator, the calendar-quarter terms would be identified from the eight tightening months alone, and the first-quarter term would drop out. The pre-period is therefore shown raw, here and in Figure A6 for the female differential. Finer calendar terms change little. With eleven month-of-year terms in place of the three quarter terms (Table A26, Panel G), τ at 22–25 is −0.038 (SE 0.010) against −0.040, and the female differential −0.070 (SE 0.011) against −0.071, so monthly seasonality moves neither estimate by more than about a fifth of a standard error under either clustering. The drift tests report whether a zero drift is rejected at two standard errors, which does not establish that the pre-period is flat.

On the paper’s window, January 2021 to November 2022, the drift at 22–25 is half a standard error from zero. At 26–30 it is positive and nearly four standard errors from zero, so that band is read against a rising pre-period, which if anything understates a later decline. The calendar midpoints of the interim and later periods are fifteen and a half months apart, and over that span the drift adds between +0.012 and +0.038 to the counterfactual. The band aged 26–30 carries less weight for that reason and because it is sensitive to industry-specific shocks. At 22–25 the interval on the slope admits a drift between about −0.018 and +0.029 over the same months. The calculation weights calendar months equally, whereas the pooled Poisson coefficient weights employer-months. An interval on a slope is also not an interval on the adjusted contrast, so the calculation illustrates scale and is not a test.

The longer window is not flat. Read within quarters, the ratio at 22–25 stood about eight log points above its 2022 level through 2019 and in the first quarter of 2020, fell to that level between the first quarters of 2020 and 2021, and stayed near it through 2023 (Table A17, upper panel). At 26–30 the same quarters show no such movement. A linear trend fitted from January 2019 is −0.0032 a month (SE 0.0004), or −0.050 over fifteen and a half months, and from January 2020 −0.0027 (SE 0.0005). At 26–30 the trend from 2019 is −0.0003 (SE

0.0003).

A backdated placebo, following [Facijs and Iacono \(2026\)](#), shows when that slope occurred. It moves every boundary of Equation (2) back in time and ends the panel in November 2022, so that no month after the launch enters. Moved back 36 months, the design compares 2021 to 2022 with 2020, and τ is -0.038 (SE 0.008). Moved back 24 months, so that it compares 2022 with 2021, τ is $+0.004$ (SE 0.007), and at 26–30 $+0.020$ (SE 0.005), the rising pre-period of that band. The interim window of the 36-month placebo is the pandemic year, and the 24-month placebo, which avoids it, returns zero. The slope from 2019 thus reflects a single movement during the pandemic year, with no drift before or after it, and the flat path from 2021 to 2023 is the counterfactual of Equation (2). Adding an indicator for March 2020 to February 2021 to the pooled drift test gives that indicator -0.020 (SE 0.005) and leaves the slope from 2019 at -0.0032 . A temporary dip would have been absorbed by the indicator, whereas a lasting step between 2020 and 2021 leaves the slope in place, as it does here.

The sex panel from January 2019 shows who carried that movement. Backdated 36 months, the female differential is $+0.052$ (SE 0.010) and the change for young men -0.064 (SE 0.009): the pandemic-year movement fell on young men at exposed employers, and young women gained relative to them. Backdated 24 months, the differential is $+0.008$ (SE 0.010). From January 2019 the differential’s linear trend is positive, $+0.0027$ a month (SE 0.0005), so the pre-period works against the negative differential of 2024 to 2025 rather than anticipating it. The raw counts on the headline employers agree. Young workers per 100 aged 31 to 69 fell between the first quarters of 2020 and 2021 by 0.55 at top-quartile employers and by 0.40 at the others, among men by 0.55 against 0.26 and among women by 0.54 against 0.52. Between the first quarters of 2023 and 2025 the same ratio for young women fell by 0.61 at top-quartile employers and did not fall at the others (8.13 and 8.14).

Clustering by industry. Exposure is built from a workforce’s occupation mix, which is correlated within industry, so an employer-clustered standard error may be too small. Table [A18](#) re-estimates every term with standard errors clustered on the employer’s three-digit industry. The coefficients are unchanged. At 22–25, τ remains significant at five per cent and the step from the tightening months does not. At 26–30 both remain significant. The standard errors of the pooled young terms roughly double, while that of the female differential rises by about a quarter, as a contrast between two sexes inside the same employer and age band implies. Table 1 of the paper reports both clusterings.

The age profile. Table 1 of the paper compares the young with their older colleagues pooled. Table [A19](#) reports every band against the prime-aged, 41–49, on the change from January 2024 against all earlier months. One band must be omitted, since the band levels are collinear with the employer-by-month effects. Figure 2 of the paper draws τ on the eight-band panel of Table [A26](#), Panel B. The gain is at the top: the band aged 50 and over gains far more than any other, 31–34 and 35–40 gain less, and on that change neither young band is distinguishable from the prime-aged. On τ , the band aged 22–25 lies at -0.024 (SE 0.011) against the prime-aged. The young thus lose ground against the prime-aged as well, by less than against all older colleagues, and the relative gain rises with age from 50–59 onwards. The pooled older reference of Table 1 therefore sits above the prime-aged, and the headline is approximately a shortfall against the prime-aged plus the gain of the oldest colleagues.

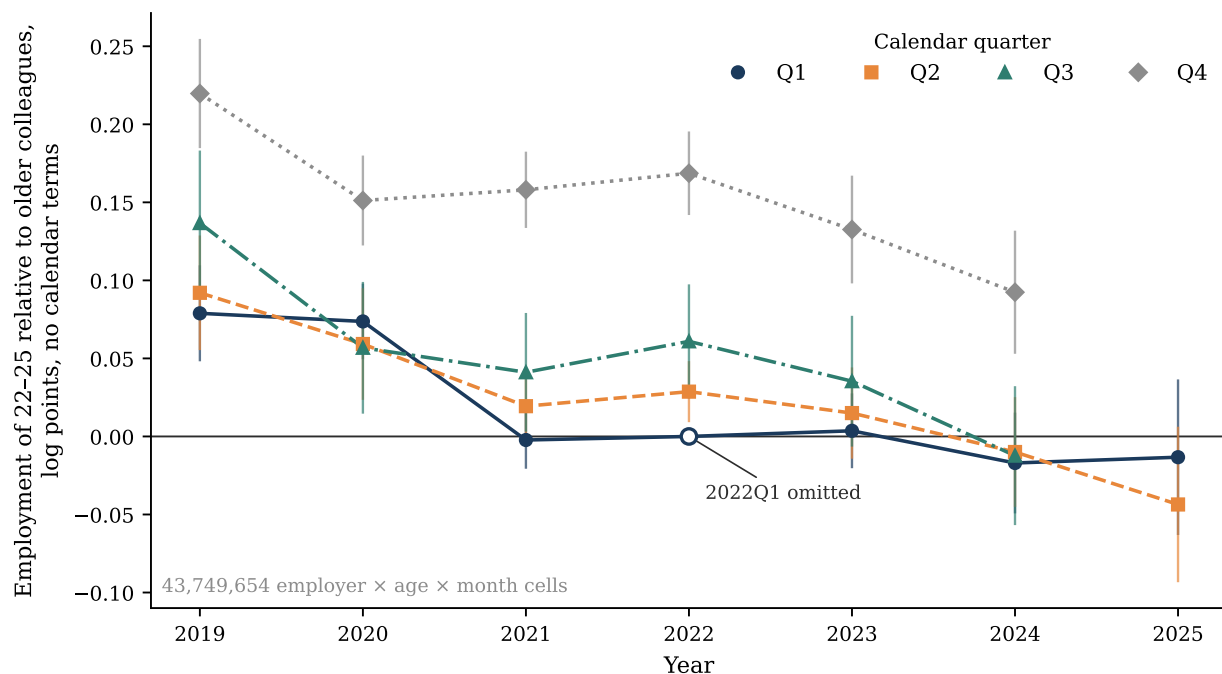


Figure A5: The quarterly path of the within-employer age gradient with no calendar terms, 2019Q1 to 2025Q2, 2022Q1 omitted. Each band is drawn as four series, one per calendar quarter, so that the cycle between quarters and the movement within a quarter across years can be read separately.

A gain at the oldest ages invites a pension-age reading: Sweden raised the earliest age for drawing the income-based pension from 61 to 62 in 2020 and to 63 in 2023, and the age to which employment protection extends from 67 to 68 and then 69 ([Socialförsäkringsutskottet, 2022](#)). The second column of Table A19 splits the oldest band at 65 in a smaller seven-band panel. Both halves gain and the rest of the profile barely moves, but the band aged 50 to 64 contains workers the reforms reach, so the split alone does not settle it. With the older reference restricted to 31–59, or to 31–49, below every pension age, τ at 22–25 is -0.035 (SE 0.010) and -0.033 (0.010), 87 and 80 per cent of the same-sample -0.040 (Table A26, Panel A). The gain is already present at 50–59 ($+0.025$, SE 0.003) and rises with age, to $+0.050$ at 60–64 and $+0.086$ at 65–69 (Panel B). The reforms may add to the gain at the oldest ages, but the young band’s decline remains when the reference group excludes every age they reach, and the gain at 50 to 59 lies below those ages.

How the young leave. Relative separation counts of the young rise from the tightening months to the later period, while the hiring step is imprecise (Table A15, Panel D). Both are counts relative to older colleagues, not hazards, and a different contrast from τ . Seniority rules protect open-ended contracts, and in 2023 more than half of employees aged 15 to 24, and six in ten of the women among them, held fixed-term ones ([Statistics Sweden, 2026a](#)). A contract that is not renewed registers here as a separation. The declarations record no contract type, so the data are consistent with this reading but cannot confirm it. An unwinding of the hiring done during the tightening would also appear as non-renewals, and the ratio ends below its level before the rate rise, although a correction need not return it exactly to that

Table A17: The quarterly path with no calendar terms, and the pre-launch drift test

Quarter	22–25	26–30
2019Q1	+0.0789* (0.0153)	−0.0034 (0.0115)
2019Q2	+0.0920* (0.0183)	−0.0016 (0.0114)
2019Q3	+0.1367* (0.0233)	+0.0042 (0.0113)
2019Q4	+0.2197* (0.0175)	+0.0481* (0.0108)
2020Q1	+0.0737* (0.0120)	−0.0054 (0.0087)
2020Q2	+0.0592* (0.0179)	−0.0071 (0.0088)
2020Q3	+0.0568* (0.0211)	−0.0147 (0.0085)
2020Q4	+0.1512* (0.0143)	+0.0206* (0.0077)
2021Q1	−0.0023 (0.0090)	−0.0219* (0.0054)
2021Q2	+0.0194 (0.0119)	−0.0111* (0.0054)
2021Q3	+0.0411* (0.0190)	−0.0080 (0.0056)
2021Q4	+0.1580* (0.0121)	+0.0407* (0.0049)
2022Q1	reference	reference
2022Q2	+0.0287* (0.0096)	+0.0070* (0.0029)
2022Q3	+0.0609* (0.0182)	+0.0137* (0.0047)
2022Q4	+0.1686* (0.0133)	+0.0607* (0.0050)
2023Q1	+0.0036 (0.0118)	+0.0049 (0.0067)
2023Q2	+0.0150 (0.0145)	+0.0043 (0.0074)
2023Q3	+0.0354 (0.0210)	+0.0027 (0.0083)
2023Q4	+0.1326* (0.0172)	+0.0427* (0.0086)
2024Q1	−0.0170 (0.0161)	−0.0112 (0.0099)
2024Q2	−0.0100 (0.0176)	−0.0148 (0.0105)
2024Q3	−0.0123 (0.0223)	−0.0225* (0.0111)
2024Q4	+0.0924* (0.0197)	+0.0155 (0.0114)
2025Q1	−0.0133 (0.0250)	−0.0599* (0.0139)
2025Q2	−0.0436 (0.0250)	−0.0785* (0.0160)
<i>The pre-launch drift test, January 2021 to November 2022</i>		
Linear trend per month	+0.00038 (0.00077)	+0.00161* (0.00044)
Over the twenty-three months	+0.0088 (0.0178)	+0.0370* (0.0100)
The tightening window	+0.0031 (0.0074)	−0.0024 (0.0034)
Zero linear drift not rejected	Yes	No
Employers	100,631	113,287
<i>The drift on longer windows, and the backdated placebo</i>		
Linear trend per month, from January 2019	−0.00321* (0.00043)	−0.00035 (0.00033)
Over 15.5 months	−0.0497* (0.0067)	−0.0054 (0.0052)
Linear trend per month, from January 2020	−0.00270* (0.00049)	
Placebo τ , boundaries moved back 36 months (2021–22 against 2020)	−0.0382* (0.0081)	
Placebo τ , boundaries moved back 24 months (2022 against 2021)	+0.0043 (0.0074)	+0.0204* (0.0046)
Employers, panels from January 2019	113,299	129,913

Notes: Upper panel: Poisson on employer-by-age-by-month counts with employer-by-month, employer-by-age and month-by-age effects and no calendar terms, 2022Q1 the reference; standard errors clustered by employer; * $p < 0.05$. Lower panel: a separate fit of a linear monthly trend on January 2021 to November 2022, with the calendar terms and the tightening window in; the second row scales it to the twenty-three months. The last row reports whether a zero trend is rejected at two standard errors, which does not establish a flat pre-period. Bottom panel: the same drift test on the pre-launch months from January 2019 and from January 2020, on panels that begin in those months; and the paper’s Equation (2) with every boundary moved back 36 or 24 months and the panel cut at November 2022, so that no month after the launch enters, τ the later-minus-interim contrast of that shifted design (for 36 months the reference is January to March 2019, where the panel begins; the 24-month “later” window holds the real tightening months).

level (Table A14).

Table A18: Every term of the headline design under two clusterings, employer and three-digit industry.

Term	Coefficient	Employer SE	Industry SE
<i>Panel A. Ages 22–25, 260 industry clusters</i>			
Tightening months ($\hat{\gamma}_1$)	+0.0159	(0.0075)*	(0.0113)
Interim, through 2023 ($\hat{\gamma}_0$)	−0.0179	(0.0097)	(0.0160)
Later period, from 2024 ($\hat{\gamma}_2$)	−0.0578	(0.0155)*	(0.0307)
Later minus interim, τ	−0.0399	(0.0102)*	(0.0191)*
Quarter 1	−0.1348	(0.0114)*	(0.0392)*
Quarter 2	−0.1305	(0.0091)*	(0.0221)*
Quarter 3	−0.1072	(0.0118)*	(0.0350)*
<i>Panel B. Ages 26–30, 263 industry clusters</i>			
Tightening months ($\hat{\gamma}_1$)	+0.0184	(0.0044)*	(0.0067)*
Interim, through 2023 ($\hat{\gamma}_0$)	−0.0080	(0.0057)	(0.0084)
Later period, from 2024 ($\hat{\gamma}_2$)	−0.0482	(0.0104)*	(0.0208)*
Later minus interim, τ	−0.0403	(0.0067)*	(0.0143)*
Quarter 1	−0.0480	(0.0035)*	(0.0115)*
Quarter 2	−0.0530	(0.0028)*	(0.0062)*
Quarter 3	−0.0439	(0.0026)*	(0.0066)*
<i>Panel C. The sexes at ages 22–25, 260 industry clusters</i>			
Young men, tightening months ($\hat{\gamma}_1$)	+0.0186	(0.0087)*	(0.0127)
Young men, interim through 2023 ($\hat{\gamma}_0$)	−0.0112	(0.0109)	(0.0172)
Young men, later period from 2024 ($\hat{\gamma}_2$)	−0.0171	(0.0179)	(0.0352)
Female differential, tightening months	−0.0057	(0.0094)	(0.0108)
Female differential, interim through 2023	−0.0144	(0.0091)	(0.0082)
Female differential, later period from 2024	−0.0858	(0.0142)*	(0.0174)*
Young women, later period from 2024	−0.1029	(0.0163)*	(0.0295)*

The headline specification of Equation (2) with standard errors clustered by employer and, alternatively, by the employer’s three-digit industry. Coefficients are identical under the two clusterings and printed once. τ is the later-period term minus the interim term. Panel C is the sex split of Equation (2) at 22–25; the row for young women is the male step plus the female differential. Stars mark $p < 0.05$ under the clustering in whose column they stand.

The sexes. The sex rows of Table 1 of the paper come from Equation (2) with every exposure-by-period term interacted with an indicator for being female and with sex-specific employer-by-age and month-by-age effects; Panel C of Table A18 prints every term. On τ , young men’s change is not distinguishable from zero (−0.006, SE 0.012), young women’s is −0.077 (SE 0.011), or 7.4 per cent, and the female differential is −0.071 (0.011), or 6.9 per cent. Both hold at one per cent when the same fit is clustered on three-digit industry. Young women were already losing ground over the thirteen months from the launch to the end of 2023, though not distinguishably so.

Before the launch the differential shows no detectable linear drift once the calendar cycle is removed (Figure A6; Table A26, Panel E). From January 2021 to November 2022 its monthly trend is +0.0014 (SE 0.0010 by employer, 0.0011 by industry), an interval from about −0.0008 to +0.0036 a month, which does not establish a flat path. Without the calendar terms the quarterly path dips in each third quarter, which a third-quarter term absorbs (−0.054, SE 0.007). With industry $\times$ age $\times$ sex $\times$ month effects the differential is −0.047 (SE 0.014, clustered by employer), 65 per cent of the same-sample −0.072: industry-

Table A19: The age profile inside exposed employers after adoption: every band against 41–49, and the oldest band split at 65.

Band, against 41–49	Six bands	Seven bands, 50 and over split at 65
22–25	−0.0192 (0.0125)	−0.0217 (0.0124)
26–30	−0.0097 (0.0086)	−0.0105 (0.0087)
31–34	+0.0181* (0.0065)	+0.0167* (0.0067)
35–40	+0.0180* (0.0053)	+0.0184* (0.0056)
41–49	reference	reference
50 and over	+0.0675* (0.0063)	—
50–64	—	+0.0392* (0.0039)
65–69	—	+0.1902* (0.0321)
Employers	153,845	104,333

Poisson with employer-by-month, employer-by-age and month-by-age effects, calendar cycle removed, one post period from January 2024, exposure the employer’s 2019 occupation mix, clustered by employer. Both columns report the change from January 2024 against all earlier months; Figure 2 of the paper draws τ on the eight-band panel of Table A26, Panel B. The six-band panel keeps every employer holding 41–49 and another band (153,845); the seven- and eight-band panels keep those holding 22–25 and another band (104,333), which is why the second column is the smaller panel. The 50–64 band contains the ages the 2020 and 2023 increases in the earliest pension age reach; the 65–69 band those the higher employment-protection age reaches. * $p < 0.05$.

specific shocks to young women absorb a third of it and do not account for it.

The differential is drawn inside the employer and the age band. Changes that move young women and young men proportionally within an employer cancel from the differential, as does their common older reference, while sex-specific employer shocks can remain. Of the three headline results, the differential lies furthest from zero, 6.6 standard errors under employer clustering and 5.3 under industry clustering, which is why the decline is reported as falling on young women.

Field of education. Fields of education differ in how exposed their graduates’ occupations are, a fact Nordström Skans and Sokolow Romin (2026) use to build a Swedish education-based exposure measure, so a difference between young workers may reflect what they studied. Education enters here only to divide the sample and never scores exposure.

Table A20 re-estimates the sex specification inside each broad track, so that young women are compared with young men holding the same broad education, each against same-track older colleagues. Weighted by young women’s shares across tracks (broad fields of education) at exposed employers, 85 per cent of the differential remains within tracks. Weighted by young men’s shares, three fifths remains. These shares are on the step from the tightening months (pooled −0.0858), not on τ (−0.071), for which no within-track split is estimated. Young women are more often in business, law and administration and less often in engineering (Table A21). The weighted within-track differentials retain 60 to 85 per cent of the pooled differential, but the comparison does not identify the contribution of educational composition. Within tracks the differential is concentrated in health, education and social care, where young men gain, and is clear in the residual fields. The tracks are broad and do not hold job content constant, so the split shows the scale of composition under two weightings, without bounding it or identifying a mechanism.

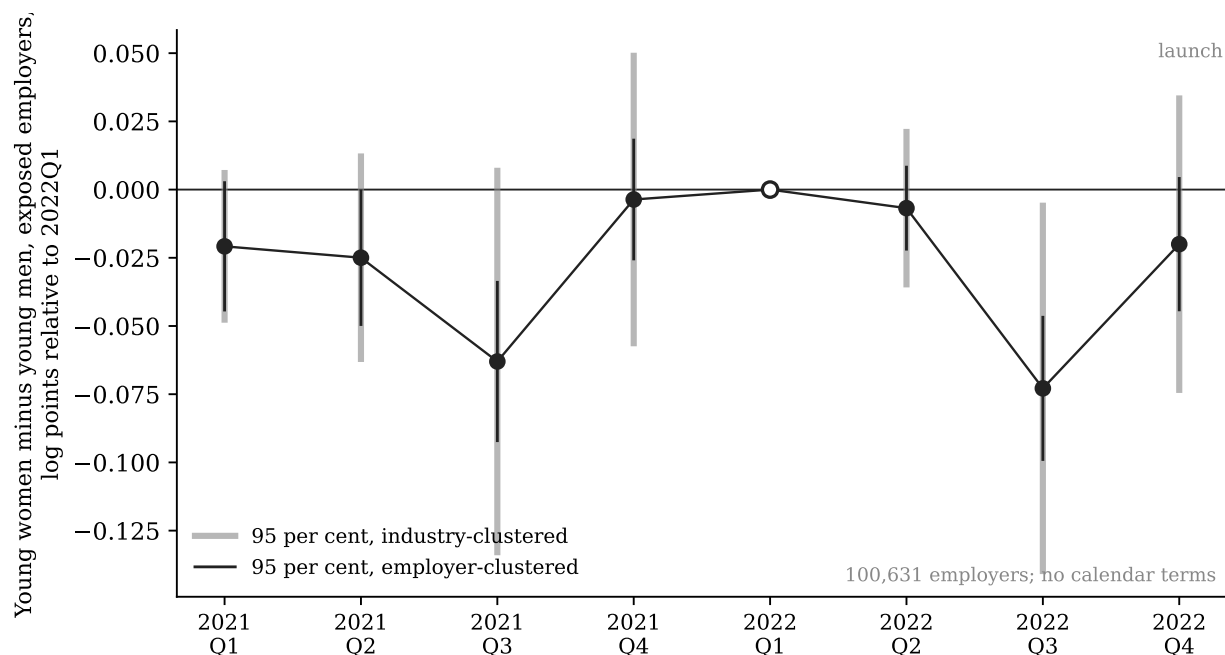


Figure A6: The female differential before the launch, raw: the quarterly coefficient on $\text{young} \times \text{exposed} \times \text{female}$, January 2021 to December 2022, against 2022Q1 (omitted, hollow), on the specification without calendar-quarter terms, so the seasonal cycle is visible. Whiskers: 95 per cent intervals clustered by employer (dark) and by three-digit industry (light). 100,631 employers. The drift test with the calendar terms on is in Table A26, Panel E.

Table A22 repeats the contrast against the prime-aged inside each track, on a three-band panel where the all-worker contrast is not distinguishable from zero at five per cent. The point estimate at 22–25 is largest in ICT and clear in business, law and administration, and not distinguishable from zero in the other three tracks. At 26–30 the ICT shortfall is deeper still. The ICT result fits Brynjolfsson et al. (2026), who report that men in their most exposed quintile are concentrated in technical work such as software development, while their pooled result on sex, steeper declines for young women, is the one that matches ours. No track is promoted above the pooled profile, and because the 2024 and 2025 education records are each worker’s 2023 record carried forward, the tracks are kept broad.

What education cannot hold constant. A track is not a job: if young women hold more exposed occupations than young men within a field, part of the differential is occupational composition, which this design cannot test without a current occupation code on every young worker. Table A23 asks the question of the national workforce instead, pricing every four-digit occupation in Statistics Sweden’s occupational register with the same index (Statistics Sweden, 2026b). Women’s occupations are the more exposed at every age, and the gap is widest at the ages the paper is about. Part of that national gap is women and men working at different employers, which the within-employer design absorbs, so the table makes occupational composition a plausible contributor to the differential without decomposing it.

An industry-specific age shock, and the credit channel. The employer-by-month effects absorb shocks that move an employer as a whole, not a shock that hits one age inside it, and

Table A20: The female differential at 22–25 within education tracks, and its weighted average.

Track	Young men	Young women minus men	Weight	Employers
All workers	−0.0171 (0.0179)	−0.0858* (0.0142)		104,217
ICT	−0.1297* (0.0593)	+0.1073 (0.0839)	0.029	6,154
Engineering, manufacturing, construction	−0.0390 (0.0252)	−0.0444 (0.0305)	0.117	51,364
Business, law, administration, social science	−0.0827* (0.0250)	−0.0408 (0.0236)	0.442	40,050
Health, education, social care	+0.1518* (0.0423)	−0.2423* (0.0409)	0.104	24,152
Other fields	+0.0011 (0.0257)	−0.0908* (0.0217)	0.301	67,917
<i>Pooled and weighted within-track estimates</i>				
Pooled differential		−0.0858* (0.0142)		
Within tracks (weighted)		−0.0731* (0.0138)		
Pooled minus weighted within-track estimate		−0.0127		
Ratio within / pooled		0.85		

The sex specification of Equation (2), with sex-specific employer-by-age and month-by-age effects, calendar cycle removed, clustered by employer, on all workers and within each broad track of the worker’s own education (two-digit SUN 2020 fields, carried forward from 2023). Exposure is the employer’s 2019 occupation mix; education only cuts the sample. Young men is the step from the tightening months; young women minus men is its female interaction. Weights are young women’s track shares in top-quartile employers in 2023. * $p < 0.05$.

Table A21: Where young women and young men work: education of employed 22–25 year olds by sex, inside and outside top-quartile employers, 2023.

	Top quartile (%)		Other quartiles (%)		Mean DAIOE score	
	Women	Men	Women	Men	Women	Men
<i>Field of education (track)</i>						
Business, law, administration, social science	44.2	29.2	19.4	12.8	67	66
Engineering, manufacturing, construction	11.7	36.4	6.5	38.4	61	53
Health, education, social care	10.4	4.0	30.9	10.4	48	47
ICT	2.9	6.8	0.6	1.2	77	76
Other fields	30.1	23.1	42.3	37.0	48	47
No record	0.5	0.4	0.2	0.2	–	–
<i>Level of education</i>						
Below upper secondary	2.0	3.7	6.1	10.0		
Upper secondary	29.2	38.1	51.2	66.6		
Post-secondary	67.1	56.3	41.3	20.3		
No record	1.7	2.0	1.4	3.1		

Shares of person-months in 2023, the last year with a contemporaneous education record, among employed workers aged 22–25. The mean DAIOE score prices the education group’s 2019 occupations for workers in top-quartile employers; it is not the exposure that defines the quartiles, which is the employer’s 2019 occupation mix. Tracks as in Table A20.

Table A24 tests the two such shocks we can name. If exposed employers cluster in an industry that stopped hiring the young for reasons unrelated to AI, the within-employer pattern would appear without any AI effect. Panel A adds three-digit industry interacted with age band and month, with the baseline refitted on the employers whose industry is known. At 22–25, 72 per cent of τ survives (−0.029, SE 0.011), as do two thirds of the step from the tightening months. At 26–30, 27 per cent of τ survives and the remainder is not distinguishable from zero, so most of that band’s decline is sectoral.

Tightening reaches employers through their balance sheets, and a leveraged employer

Table A22: The young against 41–49 inside exposed firms after adoption, by education track, calendar cycle removed.

Track	22–25 vs 41–49	26–30 vs 41–49	Employers
All workers	−0.0241 (0.0130)	−0.0090 (0.0088)	110,653
ICT	−0.1194* (0.0500)	−0.1407* (0.0350)	6,837
Engineering, manufacturing, construction	−0.0253 (0.0222)	−0.0177 (0.0158)	48,374
Business, law, administration, social science	−0.0733* (0.0215)	−0.0171 (0.0157)	37,689
Health, education, social care	−0.0251 (0.0233)	−0.0438* (0.0168)	25,379
Other fields	−0.0029 (0.0183)	−0.0421* (0.0100)	72,889

A three-band panel (22–25, 26–30 and the reference 41–49) with employer-by-month, employer-by-age and month-by-age effects, one adoption, one Riksbank and three quarter-of-year terms per young band, exposure the employer’s 2019 occupation mix, clustered by employer. Negative means the band declined relative to 41–49. Tracks as in Table A20; ICT holds 2.9 per cent of exposed employers’ young women and 6.8 per cent of their young men. * $p < 0.05$.

Table A23: How exposed to generative AI the occupations of Swedish women and men are, by age, 2024

Age	Mean exposure percentile		In the top quartile (%)	
	Women	Men	Women	Men
16–24	39.0	34.8	10.3	9.3
25–29	52.8	45.2	28.3	22.7
30–34	56.7	49.8	33.6	28.4
40–44	57.4	52.2	34.2	30.3
50–54	57.1	52.2	33.8	29.3
All 16–64	54.3	48.6	30.3	25.9

Employees aged 16 to 64 in Sweden in 2024, by four-digit occupation, age and sex, from Statistics Sweden’s occupational register (table YREG54BAS); 4,802,448 of 4,838,418 hold an occupation the DAIOE generative-AI index prices. Each column is the employment-weighted mean over the group’s occupations, and the top quartile is the index’s own. The distribution is national, so part of the gap is women and men working at different employers.

facing higher borrowing costs might cut its youngest staff first. On τ (Table A26, Panel D), the leverage terms leave the 22–25 estimate at -0.044 (SE 0.014), against -0.043 on the same 88,613 employers, and the additional term for the more leveraged half is $+0.002$ (SE 0.020), an interval from about -0.04 to $+0.04$ that admits differences of the size of τ itself. There is no detectable additional decline at the more leveraged employers, which is not the same as excluding a credit channel. At 26–30 the decline is smaller among the more leveraged exposed employers ($+0.045$, SE 0.015), evidence against the account in which higher leverage amplifies the change in age composition. The leverage term common to every employer, leverage interacted with the young band, is close to zero at 22–25 ($+0.002$, SE 0.007) and negative at 26–30 (-0.010 , SE 0.005), so what credit channel the data show reaches the older young band. Only limited companies file balance sheets. Their step is larger than that of the full panel, so the step among public-sector and unincorporated employers, which this test excludes, is smaller. Leverage is a proxy for exposure to the rate cycle, not a measure of it.

Panel B of Table A24 reports a simpler version of the test: the step against April 2022

to December 2023, without interim or calendar terms, among the employers that file a 2019 balance sheet, split at median leverage. The additional step for the more leveraged half is +0.032 (SE 0.027).

Table A24: An industry-specific age shock and the credit channel as rival explanations for the decline of the young inside exposed employers.

	22–25	26–30
<i>Panel A. An industry-specific age shock</i>		
τ , baseline on the same employers	−0.0400* (0.0103)	−0.0403* (0.0068)
τ with industry $\times$ age $\times$ month	−0.0287* (0.0114)	−0.0108 (0.0070)
Retained, τ (per cent)	72	27
Step from the tightening months, baseline	−0.0579* (0.0155)	−0.0483* (0.0104)
Step from the tightening months, with industry $\times$ age $\times$ month	−0.0388* (0.0143)	−0.0152 (0.0098)
Retained, step (per cent)	67	31
Employers	103,728	116,378
Three-digit industry groups	259	262
<i>Panel B. The credit channel, employers with a 2019 balance sheet</i>		
Baseline on this sample	−0.0887* (0.0116)	−0.0572* (0.0088)
Baseline, full panel	−0.0658* (0.0125)	−0.0498* (0.0080)
Step at January 2024, less leveraged half	−0.1062* (0.0174)	−0.0925* (0.0147)
Additional step, more leveraged half	+0.0319 (0.0267)	+0.0650* (0.0200)
Step at January 2024 averaged over the halves	−0.0903* (0.0115)	−0.0600* (0.0089)
Leverage $\times$ young, all employers in the sample	−0.0075 (0.0088)	−0.0186* (0.0072)
Retained, averaged step over baseline (per cent)	102	105

Poisson with the paper’s three fixed effects, exposure the employer’s 2019 occupation mix, clustered by employer. Panel A carries the interim and calendar-quarter terms of Equation (2), so both τ and the step from the tightening months are reported. Panel B carries neither: its step is at January 2024 against the April 2022 to December 2023 average, without the calendar cycle removed, so each row reads against the baseline in its own panel. Panel A: industry is the employer’s three-digit NACE in 2019 (under 2 per cent coded from another year), interacted with age band and month, on the employers that carry a code. Panel B: leverage is one minus equity over assets on the 2019 balance sheet, split at the median, for the limited companies that file one (85 per cent of each panel); the averaged step is the first plus half the second credit row. * $p < 0.05$.

The payroll reduction for young workers. A third rival is specific to the young. Sweden reduced the employer social contribution for young workers from 1 January 2021 to 31 March 2023 ([Sveriges riksdag, 2021](#)). The reduction applied to those who at the start of the year had turned 18 but not 23. It lowered the rate to roughly 19.7 per cent, against the standard 31.4, on monthly pay up to 25,000 kronor, and to the retirement-pension contribution alone in June to August 2021 and 2022. Its withdrawal raised the cost of the youngest workers inside the window. Eligibility ran by birth year, so in 2023 it covered those born 2000 to 2004 ([Skatteverket, 2023](#)), and a worker born in 2000 is 24 in 2024. The reduction was common to every employer, so it threatens the design only if its effect on the young-to-older ratio differs with exposure, which can happen if the share of covered cohorts within the band differs with exposure.

Table A26, Panel C, follows fixed birth cohorts, so that no one enters or leaves a group with age. Both the cohorts the reduction never covered, born 1994–1997, and those it

did, born 1998–2003, grow at exposed employers relative to the reference cohorts (+0.054, SE 0.007, and +0.215, SE 0.014), and the gradient rises with statutory years of eligibility. Those cohorts were finishing their education and entering employment over the window. The estimates therefore combine eligibility with age-related changes in employment and in sorting across employers. Because eligibility falls with birth year, the number of eligible years measures age as much as eligibility. The exercise does not isolate the payroll reduction, and it does not bear on the headline, which compares whoever is 22–25 in each month.

A split of 22–25 by age does not separate covered from uncovered workers either. Fitted separately against the same older colleagues, τ is -0.026 (SE 0.012) for those aged 22 and 23 and -0.050 (0.011) for those aged 24 and 25, and most workers in both halves were covered at some point between 2021 and March 2023. The two fits share their comparison bands, so no test of the difference is reported, and the halves differ in more than eligibility, since students and first entrants weigh more at 22 and 23.

Remote work. Remote work is a rival reading of the oldest band’s gain: [Lambert and Schindler \(2026\)](#) argue that it raises the cost of supervising and developing juniors. The headline design takes it up directly. The employer’s teleworkability, standardised with incumbent weights, enters beside the exposure indicator on the same 104,217 employers, interacted with the young band and the same period and calendar terms. τ on exposure is then -0.053 (SE 0.018), against -0.040 without it, and the teleworkability term is $+0.008$ per standard deviation (SE 0.006). On its own, teleworkability gives -0.007 (SE 0.003). The female differential on exposure is -0.059 (SE 0.019) with teleworkability in, against -0.071 without, and its teleworkability term is -0.007 (SE 0.008). The two scores correlate at 0.88, so the joint estimates are less precise than either alone. Split samples do not discriminate: among employers in the lower half of teleworkability only 838 are in the top exposure quartile, and τ there is $+0.020$ (SE 0.063); among those in the lower half of exposure, 439 are in the top teleworkability quartile, with -0.025 (SE 0.053).

A second check, in Table [A25](#), scores occupations continuously instead of by quartile, so that exposure varies inside an employer-month and the band levels are identified. In this check each age band is scored from the 2019 occupations of its own members, so workers aged 22–25 are classified. The headline design never classifies them, and the table is a check on the measure rather than a second estimate. At 22–25 the two AI scores return the same coefficient in this specification, -0.013 per standard deviation of DAIOE and of the Eloundou rating (SE 0.011 and 0.009). The headline design itself is re-estimated on the Eloundou classification in Table [A26](#), Panel H. Employment at 41–49 declines on both AI measures and not on teleworkability, but the gain of the oldest band is at least as large on teleworkability as on DAIOE, so on this check it is not specific to AI. Entered together, the two scores separate at the young bands, where they take opposite signs: at 22–25 the loading is -0.032 (SE 0.014) per standard deviation of AI exposure and $+0.027$ (0.012) per standard deviation of teleworkability, and at 50 and over neither is distinguishable from zero.

The exposure specification. The paper treats the top quartile of the employer score against the other three. Measured continuously, τ at 22–25 is -0.0105 (SE 0.0040) per baseline standard deviation of the score, a relative decline of about one per cent per standard deviation. Entered as quartiles against the first, the second and third are not distinguishable

Table A25: The age profile on a continuous occupation-scaled exposure measure.

Age band	DAIOE	Eloundou	Teleworkable
22–25	−0.0132 (0.0112)	−0.0132 (0.0085)	+0.0069 (0.0101)
26–30	−0.0016 (0.0077)	−0.0054 (0.0061)	+0.0051 (0.0069)
31–34	+0.0039 (0.0073)	−0.0028 (0.0057)	+0.0063 (0.0065)
35–40	−0.0028 (0.0067)	−0.0106 (0.0055)	+0.0040 (0.0060)
41–49	−0.0208* (0.0073)	−0.0311* (0.0059)	−0.0034 (0.0068)
50+	+0.0316* (0.0110)	+0.0152 (0.0101)	+0.0379* (0.0096)

Poisson with employer-by-month, employer-by-age and month-by-age effects, one post period from January 2024, clustered by employer; no calendar terms. Occupations are scored continuously, so exposure varies within an employer-month, and each coefficient is per standard deviation of that measure. The Eloundou column is the β rating of [Eloundou et al. \(2024\)](#); the teleworkable column is the classification of [Dingel and Neiman \(2020\)](#); both reach SSYK by the crosswalk chain of Section VI.1. A check on the measure, not a second estimate of the step. * $p < 0.05$.

from it (+0.002, SE 0.007, and +0.009, SE 0.017) and the top quartile declines (−0.037, SE 0.010; Table A26, Panel F). The decline is concentrated among the most exposed employers. The quartile estimates support the top-quartile comparison without establishing a threshold, since the second and third quartiles are imprecise rather than zero and the third is not contrasted with the fourth directly.

Reclassifying employers by the GPT-exposure rating of [Eloundou et al. \(2024\)](#) leaves the female differential and the estimate at 26–30 almost unchanged, but gives a smaller, less precise pooled estimate at 22–25 (Table A26, Panel H). The construction is otherwise the paper’s: the same crosswalks, the 2019 codes of incumbents aged 31 to 69, the three-digit scoring rule, the incumbent floor and the employment-weighted quartile cut, each index keeping its own classification. The comparisons use the 103,064 employers both indices score at 22–25 and the 115,628 at 26–30. Of DAIOE’s top-quartile employers, 92.5 per cent are in the Eloundou top quartile too (91.8 per cent by incumbent employment), and the employer scores have a Spearman correlation of 0.91. The female differential is −0.071 (SE 0.012), against −0.072 (0.011) on DAIOE. The estimates at 26–30 are −0.039 on both (SE 0.007). At 22–25 the Eloundou classification gives −0.024 (SE 0.019 by employer, 0.023 by industry), against −0.039 (SE 0.010) on DAIOE. Its employer-clustered 95 per cent confidence interval, [−0.062, +0.013], includes both zero and the DAIOE estimate. The separate covariance matrices do not supply the cross-model covariance a test of the difference would need.

Table A27 asks which of the two rankings picks out the employers that use AI, on the surveys of Figure A4 and the employers both indices score. Taken one at a time, the two classifications predict reported use about equally. Entered together, DAIOE carries the prediction of AI use before ChatGPT, in the 2019 surveys and in 2021 (a difference of 10.7 points, SE 5.0, on any AI in 2021, and 9.4, SE 1.6, in the business survey of research and development), and of workers’ reported use of generative AI in 2024 (27.9 points against −6.2, a difference of 34.1, SE 11.0). In the 2023 firm survey the two cannot be told apart (3.4, SE 5.5, on any AI; −0.9, SE 3.2, on language generation), and among public-sector organisations the Eloundou ranking fits better but imprecisely. Where the two rankings disagree, DAIOE’s top quartile is therefore the one more closely associated with reported AI use in most of the surveys.

Table A26: Robustness of τ : pension ages, birth cohorts, credit, the female differential, the exposure specification, monthly seasonality and the exposure index

<i>Panel A. τ with the older reference restricted, common sample</i>			
Reference band	31–69	31–59	31–49
22–25 (98,508 employers)	−0.0405 (0.0103)	−0.0350 (0.0103)	−0.0325 (0.0101)
26–30 (112,114 employers)	−0.0407 (0.0068)	−0.0354 (0.0067)	−0.0308 (0.0062)
<i>Panel B. τ for each band against 41–49, one panel of 104,333 employers</i>			
22–25	−0.0239 (0.0110)	50–59	+0.0254 (0.0034)
26–30	−0.0174 (0.0077)	60–64	+0.0504 (0.0059)
31–34	+0.0108 (0.0058)	65–69	+0.0857 (0.0175)
35–40	+0.0154 (0.0049)		
<i>Panel C. Fixed birth cohorts against those born 1956–1990, later minus interim, 128,286 employers</i>			
Never covered by the payroll reduction, born 1994–1997			+0.0536 (0.0068)
Ever covered, born 1998–2003			+0.2155 (0.0139)
Difference			+0.1618 (0.0113)
Per statutory year of eligibility, born 1994–2003			+0.0852 (0.0061)
<i>Panel D. The credit test on τ, employers with a 2019 balance sheet</i>			
	22–25	26–30	
Same-sample τ	−0.0429 (0.0102)	−0.0415 (0.0076)	
τ , less leveraged half	−0.0439 (0.0143)	−0.0658 (0.0126)	
Additional τ , more leveraged half	+0.0020 (0.0201)	+0.0450 (0.0154)	
Young $\times$ leverage, all employers	+0.0024 (0.0071)	−0.0100 (0.0053)	
Employers	88,613	98,669	
<i>Panel E. The female differential at 22–25</i>			
Same-sample τ , 103,728 employers with an industry code			−0.0716 (0.0109)
With industry $\times$ age $\times$ sex $\times$ month effects			−0.0467 (0.0141)
Pre-launch drift per month, Jan 2021 to Nov 2022, by employer			+0.00137 (0.00097)
the same, clustered by industry			+0.00137 (0.00112)
<i>Panel F. The exposure specification at 22–25, 104,217 employers</i>			
Continuous score, per baseline standard deviation			−0.0105 (0.0040)
Second quartile against the first			+0.0015 (0.0074)
Third quartile against the first			+0.0085 (0.0166)
Top quartile against the first			−0.0367 (0.0103)
<i>Panel G. Eleven month-of-year terms in place of the three calendar-quarter terms, 104,217 employers</i>			
	Quarter terms (Table 1)	Month-of-year terms	
τ at 22–25, SE by employer and by industry	−0.0399 (0.0102; 0.0191)	−0.0378 (0.0103; 0.0192)	
Female differential at 22–25, SE by employer and by industry	−0.0714 (0.0109; 0.0136)	−0.0704 (0.0109; 0.0135)	
<i>Panel H. Employers classified by the Eloundou et al. (2024) rating in place of DAIOE, on the employers both indices score</i>			
	DAIOE, common employers	Eloundou rating	
τ at 22–25, 103,064 employers	−0.0391 (0.0103)	−0.0243 (0.0191; 0.0226)	
Female differential at 22–25, 103,064 employers	−0.0724 (0.0109)	−0.0710 (0.0121; 0.0138)	
τ at 26–30, 115,628 employers	−0.0395 (0.0068)	−0.0393 (0.0067)	

τ is the later-period coefficient minus the interim one from the same Poisson fit, standard errors clustered by employer unless stated. Every fit reproduces Table 1 of the paper on its own sample before any check is run. Panel A: the young band against the stated older bands, on the employers that hold both the young band and ages 31–49. Panel B: eight bands, seven contrasts against 41–49, in one fit. Panel C: groups defined by birth year alone, so no one enters or leaves a group as they age; the reduction covered pay from January 2021 to March 2023 for those who had turned 18 but not 23 at the start of the year (Sveriges riksdag, 2021). Panel D: leverage is one minus equity over total assets in the 2019 balance sheet, split at the median (0.693 at 22–25, 0.689 at 26–30); the rows are the terms of one fit. Panel E: the drift is the monthly trend on the young $\times$ exposed $\times$ female term with calendar-quarter terms; a third-quarter term absorbs a seasonal dip (−0.054, SE 0.007). Panel F: the score’s baseline standard deviation is 17.6 percentile points. Panel G: the month-of-year terms, December omitted, are interacted with exposure and the young band (and with sex on the sex panel), everything else as in Equation (2); the two columns come from separate fits, so the difference between them carries no standard error of its own. Panel H: employers re-scored through the same chain (2019 codes of incumbents aged 31–69, three-digit book, incumbent floor, employment-weighted quartile cut) with the Eloundou rating as the occupation score, each index keeping its own top quartile; 92.5 per cent of DAIOE’s top-quartile employers (91.8 per cent by incumbent employment) are in the Eloundou top quartile, and the employer scores correlate at 0.91 (Spearman); the Eloundou cells give the employer-clustered standard error and, at 22–25, the industry-clustered one; separate fits, so the differences carry no standard error.

Estimation details. Quartiles are weighted by incumbent employment, so each holds a quarter of the 2019 incumbent person-months rather than a quarter of employers. The bottom and top quartiles hold many small employers and the middle two fewer, larger ones. Age is taken from birth year, so a worker’s band changes every January, which the month-by-age effects absorb, and the quarterly path shows no step at any January. Standard errors are

Table A27: Reported AI use and the two exposure rankings: top-quartile differentials on each classification alone and on both together

Survey and outcome	Separately		Together			<i>N</i>
	DAIOE	Eloundou	DAIOE	Eloundou	Difference	
ICT survey 2019, any AI	+18.1 (1.2)	+16.9 (1.2)	+13.5 (2.4)	+5.1 (2.3)	+8.3 (4.5)	3,805
IT-expenditure survey 2019, any AI	+5.0 (1.3)	+4.0 (1.3)	+7.2 (2.8)	−2.4 (2.8)	+9.5 (5.4)	3,818
R&D survey, business, 2019–21, AI in R&D	+5.5 (0.4)	+4.4 (0.4)	+7.3 (0.8)	−2.1 (0.8)	+9.4 (1.6)	12,508
R&D survey, public sector, 2019–21, AI in R&D	+29.4 (3.8)	+30.9 (3.8)	+3.6 (10.6)	+27.5 (10.7)	−23.8 (20.9)	1,006
ICT survey 2021, any AI	+18.7 (1.3)	+17.2 (1.3)	+14.9 (2.6)	+4.2 (2.5)	+10.7 (5.0)	4,141
ICT survey 2021, language generation	+7.6 (0.8)	+6.9 (0.8)	+6.5 (1.3)	+1.2 (1.2)	+5.3 (2.5)	4,141
ICT survey 2023, any AI	+20.7 (1.5)	+20.0 (1.4)	+12.6 (2.9)	+9.2 (2.8)	+3.4 (5.5)	3,582
ICT survey 2023, language generation	+10.5 (1.0)	+10.5 (1.0)	+5.2 (1.7)	+6.1 (1.7)	−0.9 (3.2)	3,582
Worker survey 2024, generative AI	+22.3 (2.5)	+18.9 (2.5)	+27.9 (5.7)	−6.2 (5.6)	+34.1 (11.0)	2,274

Top-quartile differential in reported use, percentage points, with robust standard errors, on the employers both indices score. Separately: each classification’s top-quartile indicator alone. Together: both indicators in one regression; the difference is DAIOE minus Eloundou, its standard error from the same fit. Firm surveys: linear probability models with log 2019 employment; *N* firms. Worker survey: survey-weighted, *N* respondents linked to one employer. The two top quartiles share most employers, so the joint fit is identified from those on which they disagree.

clustered by employer, the level at which exposure is assigned.

III.3. The precision of the employer score

Table A28 asks whether the estimate depends on how precisely each employer is scored. An employer’s exposure is an average over its incumbents aged 31 to 69 in 2019, and at the median employer it rests on two people. Classical misclassification of a binary exposure indicator would bias the estimate towards zero. A noisy score also has fatter tails, however, so small employers are over-represented at both ends of the exposure distribution, which changes the composition of the exposed group.

Panel A reports reliability, the share of the variance in a firm’s score that is signal rather than sampling noise, at the quantiles of the incumbent count, first across employers and then across employment. It is model-based: the employment-weighted variance of the worker-level score is split into a between-employer part σ_B^2 , netted of its own sampling noise, and a within-employer part σ_W^2 , and an employer scored from n incumbents has reliability $\lambda(n) = \sigma_B^2 / (\sigma_B^2 + \sigma_W^2/n)$. Two employers in five have a score whose reliability is below one half, but they are small and together hold three per cent of incumbent employment. The median worker is at an employer whose score rests on hundreds of colleagues and has a reliability close to one.

Panel B raises the qualifying floor twelvefold, from one worker for five months to five workers for a full year, which halves the panel and removes the employers whose scores are thinnest, with the national quartile carried over unchanged. τ barely moves at either band (-0.041 at both). Split at four incumbents, τ is concentrated in employers scored from four or more incumbents (-0.042 at 22–25, -0.041 at 26–30). In employers scored from one to three incumbents it is smaller and imprecise at 22–25 (-0.019 , SE 0.017) and zero at 26–30. The step from the tightening months is similar in the two size groups at 22–25 because the smallest employers’ interim coefficient is already negative. A score resting on one to three incumbents is noisy, and the smaller estimate there may reflect scoring error or differences between small and large employers.

Table A28: The precision of a employer score averaged over its incumbents, and what it does to the estimates

<i>Panel A. Reliability of the employer score</i>		
	Incumbents	Reliability
10th percentile of employers	1	0.43
25th percentile of employers	1	0.43
50th percentile of employers	2	0.60
75th percentile of employers	5	0.79
90th percentile of employers	13	0.91
10th percentile of incumbent employment	5	0.791
25th percentile of incumbent employment	30	0.958
50th percentile of incumbent employment	645	0.998
75th percentile of incumbent employment	5,210	1.000
90th percentile of incumbent employment	13,527	1.000
Below a reliability of one half	40.6% of employers	3.0% of employment

<i>Panel B. The estimates by the number of incumbents behind the score</i>		
	Ages 22–25	Ages 26–30
Reported floor, five incumbent person-months, τ	−0.0399 (0.0102)	−0.0403 (0.0067)
step from the tightening months	−0.0578 (0.0155)	−0.0482 (0.0104)
employers in the panel	104,217	117,090
Floor of sixty person-months, τ	−0.0409 (0.0112)	−0.0406 (0.0072)
step from the tightening months	−0.0537 (0.0169)	−0.0498 (0.0110)
employers in the panel	52,333	57,113
Employers with 1 to 3 incumbents, τ	−0.0188 (0.0167)	−0.0003 (0.0141)
step from the tightening months	−0.0578 (0.0223)	+0.0286 (0.0183)
employers in the panel	39,215	45,195
Employers with 4 or more incumbents, τ	−0.0419 (0.0109)	−0.0410 (0.0070)
step from the tightening months	−0.0569 (0.0165)	−0.0499 (0.0108)
employers in the panel	65,002	71,895

Panel A: reliability is the share of the variance in the employer score that is signal rather than sampling noise, $\lambda(n) = \sigma_B^2/(\sigma_B^2 + \sigma_W^2/n)$ for an employer scored from n incumbents, with the between-employer variance netted of its own sampling noise; it rises with the number of incumbents behind the score and is reported, not used to correct any estimate, since the measurement error is not classical. Panel B: τ and the step from the tightening months of Equation (2), clustered by employer, with the national exposure quartile carried into every split unchanged. The size groups split at four incumbents, because the incumbent count is too skewed for thirds.

III.4. Military conscription

Sweden decided to resume conscription in March 2017, the first conscripts began basic training in January 2018, and the plan raises the annual intake from 5,000 in 2019 to 8,000 in 2025; those selected are called up at 18 (see [Plikt- och prövningsverket, 2025](#)). A growing intake reduces the measured civilian employment of the youngest cohorts, which could account for part of the raw fall in young headcount.

Its common age-specific effects are absorbed by the month-by-age effects, and the intake is drawn from across the occupational distribution rather than from employers of one exposure level, although exposure-related differences in entry and return to employment could remain. The band the paper reports begins at 22, above the conscription ages, as in [Brynjolfsson et al. \(2026\)](#), although service delays university entry, so conscripted cohorts re-enter the

labour market inside the 22–25 band and partly inside 26–30. It remains a reason not to read the raw fall in young headcount as a labour-demand statement.

III.5. Comparison with related studies

Table A30 compares the study with the estimates of whether generative-AI exposure affects employment or postings differentially by age or seniority; the studies differ in data, country, exposure measure, design and whether age is tested directly. The table carries the samples and coefficients, and the text below draws the comparisons that bear on the Swedish result.

The original US payroll study finds employment of workers aged 22 to 25 in exposed occupations falling behind that of less exposed peers of the same age, through reduced hiring of the young and persisting when exposure to interest-rate increases and remote work is controlled for. Its authors present these patterns as descriptive facts (Brynjolfsson et al., 2026). Among the estimated designs, US and UK studies using résumé and profile data find negative effects on junior workers (Hosseini Maasoum and Lichtinger, 2026; Klein Teeselink, 2025), and Hosseini Maasoum and Lichtinger (2026) frame generative AI as seniority-biased technological change inside adopting firms, while in US postings the decline in substitutable occupations falls more on positions requiring a degree and experience (Liu et al., 2026).

Nordic register studies are mixed: the Finnish analyses report nulls at occupation level (Kauhanen and Rouvinen, 2025) and under the firm-level Poisson design (Kauhanen and Rouvinen, 2026), the Danish survey-register study finds no change in adopting workplaces’ employment share of workers aged 22 to 25, alongside precise nulls on earnings and hours while employers reorganise tasks (Humlum and Vestergaard, 2025), and the Danish adoption study finds a decline concentrated on the young (Bonin et al., 2026). Two further US readings sit outside the table, which lists working papers and articles: Massenkoff and McCrory (2026) use the monthly panel of the Current Population Survey to measure the share of workers aged 22 to 25 who start a job in occupations in the top quartile of their usage-based exposure measure, against occupations with no observed exposure; in a difference-in-differences framework the post-ChatGPT rate is 14 per cent below its 2022 level, which they describe as just barely statistically significant, with no such decrease above 25 and no differential rise in unemployment, while Iscenko and Curto Millet (2026) attribute the US entry-level pattern to monetary policy rather than AI.² Timing alone does not exclude delayed monetary effects, which is why the credit channel is tested directly (Section III.2).

The most direct contrast is with Kauhanen and Rouvinen (2026), who replicate the Brynjolfsson firm-level Poisson design on population-level data and find no displacement among young Finnish workers. The Finnish study compares occupational exposure groups

²The most recent studies are not tabulated. Westby et al. (2026) find junior software-developer vacancies, those asking for three years’ experience or less, falling 14 to 15 per cent relative to senior ones; Orr et al. (2026) find that, relative to 2022 and to majors in the bottom six deciles, the probability of employment in the quarter after graduation fell by five percentage points for bachelor’s graduates of the most AI-exposed decile of majors, up to half of which may reflect moves into self-employment and graduate study; Hernæs and Kostøl (2026) estimate the relative change of young Norwegian workers in the most exposed occupational quintile as an imprecise zero; and in Lambert and Schindler (2026) the generative-AI effect on junior hiring across four countries attenuates sharply once remote-work exposure is estimated jointly.

within age groups and firms; we compare age groups within employers classified by their 2019 exposure, so a shared estimator does not make the two estimands equivalent. Three further differences may matter. The Finnish sample is private employers with at least ten employees in each age group in each month and at least 100 across the exposure quantiles, where ours adds the public sector and part-time workers with a floor of five incumbent person-months. The Finnish study rates occupations with the Eloundou rating, among others, and reclassifying our employers by that rating leaves the female differential and the estimate at 26–30 almost unchanged but gives a smaller, less precise pooled estimate at 22–25 (Table A26, Panel H). We also checked which ranking better picks out employers that use AI: where the two disagree, DAIOE’s top quartile is the more closely associated with reported AI use in most of the Swedish surveys (Table A27). The same check on Finnish data would show whether the rankings diverge in the same way there. Finally, a design dated from the launch counts 2023, when the decline had only begun, as part of its post-launch period, so a Swedish design dated in that way would also report a smaller effect. Institutions may matter too (Table A29), and none of this can be adjudicated from outside the Finnish data.

[Facijs and Iacono \(2026\)](#) estimate four designs on Norwegian population registers under two exposure indices. Their estimates for ages 22–25 are negative in each of the three firm and occupation designs and significant in none (-0.070 , SE 0.066 , within firms). The two firm designs identify from fewer than a hundred firms, where the Swedish panel holds 104,217 employers at 22–25, and their pre-window, from January 2015, is several times the length of ours. Their synthetic difference-in-differences also carries a backdating placebo: a false treatment date in the second half of 2019 produces larger absolute estimates than the real date in most age bands, which they read as post-2022 dynamics partly continuing a secular trend. We ran the same kind of placebo on our design (Section III.2, the pre-period). Moved back 24 months, so that it compares 2022 with 2021, it returns zero ($+0.004$, SE 0.007). Moved back 36 months, it picks up the one earlier movement, in the pandemic year (-0.038 , SE 0.008), and in the two years before the launch the drift at 22–25 lies within two standard errors of zero. Our estimate is therefore not the continuation of a trend running since 2021, although the pandemic year shows that a shock reaching young workers at exposed employers does register in this ratio.

[Bonin et al. \(2026\)](#) treat adoption itself as the treatment, following Danish firms that first used AI in 2023 against firms reporting no use, within industry-month cells and against each firm’s own trend. Adopters fall about 6 per cent further below trend than non-adopters after two years and about 11 per cent by late 2025, a gap carried by adopters with fewer than 100 full-time equivalents, and workers under 30 are about 20 per cent below trend against 5 per cent for older workers. Their adjustment runs through hiring, the stock of workers with under two years of tenure falling about 28 log points below trend while employment of high-tenure workers rises, where ours shows separations of the young rising and hiring imprecise. Denmark has among the lowest employment protection in the OECD ([Bonin et al., 2026](#)), while Sweden protects open-ended contracts by seniority and in 2023 more than half of its employees aged 15 to 24 held fixed-term contracts ([Statistics Sweden, 2026a](#)).

Employment protection differs across the four countries (Table A29). On the OECD indicators, Sweden combines the strictest protection of regular contracts among the four with the least regulated temporary contracts, and has held both positions since 2015 ([OECD](#),

2026). Danish regular contracts are the least protected, and Norway regulates temporary contracts most strictly. Where permanent contracts are costly to end and temporary ones are not, adjustment can fall on the young through non-renewal, which registers as a separation, as in the Swedish flows.

Table A29: Strictness of employment protection, Nordic countries

	Regular contracts		Temporary contracts	
	2022	2025	2022	2025
Sweden	2.48	2.43	1.15	1.24
Finland	2.37	2.37	1.42	1.42
Norway	2.27	2.27	2.28	2.53
Denmark	1.84	1.84	1.56	1.56

Notes: OECD indicators of employment protection, version 4, regulation in force on 1 January; scale 0 to 6, higher is stricter. Regular contracts: individual dismissals.

Source: OECD (2026).

Near-universe US flow data point the same way. Tucker (2026) finds hires of workers aged 22 to 24 falling immediately after ChatGPT in the most exposed quintile of industry-by-state cells, with regression-adjusted early-career employment 12 per cent lower ten quarters on and monetary shocks accounting for up to a quarter of that gap but not for the fall in hires. The design compares industries rather than employers, and part of such gradients may reflect composition across industries (Bonin et al., 2026), which the within-employer comparison holds fixed.

The first international estimates come from Chandar and Klein Teeselink (2026), who measure adoption in 41 countries in Revelio Labs data and match treated foreign affiliates of multinational groups to controls within country and sector, instrumenting adoption with the adoption rate among other multinational parents headquartered in the same metropolitan area. Junior is Revelio seniority levels 1 and 2 against 3 to 7 for senior, where ours is a cut by age. At their March 2026 endpoint the junior share of employment falls by 0.019 (SE 0.004), log junior employment by an insignificant 0.025 (0.023), and log senior employment rises by 0.067 (0.022): in their reading the decline in the junior share “primarily reflects growth in senior employment rather than a fall in junior employment”.

That is the shape of the Swedish profile, in which the young fall most relative to their oldest colleagues and the relative gain rises with age (Table A26, Panel B). On wages, Azar et al. (2026) find declines within junior positions across age bands in the United States. Their sample is not ours, and they say so: professional profiles over-represent managers and professionals, and the authors describe their estimates as relevant to digitally attached multinational firms. Across their countries junior losses are weakly smaller where labour regulation is more rigid ($r = +0.20$, not significant), and Sweden, comparatively rigid, nonetheless shows the relative employment of young workers falling and separations of the young rising.

The Swedish result is therefore not an outlier in its shape: where the international evidence finds a junior loss, it finds it as a widening gap against senior colleagues. It differs in its population, a source that covers every employee with a payslip in Sweden, in every

sector and with part-time workers, where the other Nordic register designs are private-sector and the multinational evidence rests on public professional profiles. The Nordic disagreement remains open: the estimator does not explain it, and the sample, the index, the dating and institutions remain candidates that only the Finnish and Norwegian data can settle. For surveys of this literature see [Chandar \(2025\)](#) and [Kolko \(2026\)](#).

Table A30: Studies of whether AI exposure affects employment or job postings differentially by age or seniority: the original descriptive study (first row) and estimated designs. ES = event study; DiD = difference-in-differences; SDiD = synthetic DiD. Not listed: wage studies such as [Azar et al. \(2026\)](#), commentaries and research notes, and the most recent studies (footnote to the text).

Study	Country	Data	Method	Estimand or comparison	Coverage	Age tested?	Key result
Brynjolfsson et al. (2026)	US	ADP payroll	Descriptive; within-firm Poisson ES as a check	Exposed against less exposed occupations within an age group, inside firms	Clients of one payroll processor; 3.5–5 million workers a month in the panel	Yes	Ages 22–25 about 19% below less exposed peers of the same age by mid-2026, descriptively; through hiring
Hosseini Maa-soum and Lichtinger (2026)	US	Résumés + postings	Firm DiD and triple difference	Junior against senior employment, adopting against non-adopting firms	Online résumés, 65 million workers at more than 280,000 firms	Yes (seniority)	Junior employment about –9%; senior broadly unchanged; slower hiring, not separations
Klein Teeselink (2025)	UK	Revelio Labs profiles + postings	DiD	Firms and occupations by LLM exposure; junior against senior positions	Public professional profiles, firms with at least 100 employees	Yes (seniority)	Junior employment –0.4% per SD of exposure, senior not significant; posting probability –1.0 pp
Liu et al. (2026)	US	Lightcast postings	DiD + ES	Postings, high against low AI-substitution occupations at similar exposure	285 million online postings, 2018–25	No (posting requirements)	Postings –9% on average, –15% by year 3; postings requiring a degree and experience fall more
Facijs and Iacono (2026)	Norway	Population registers, private sector	Within-firm DiD, shift-share, SDiD	Exposure groups within an age group, inside firms, and others	Private firms, full-time main jobs, ages 18–66	Yes	Negative at 22–25 in all three firm and occupation designs, significant in none; firm designs identify from 45 and 82 firms
Kauhanen and Rouvinen (2026)	Finland	Income register, private sector	Poisson ES	Exposure groups within an age group, inside firms	Private employers with 10 or more employees per age group and month, 100 in total	Yes	Null; if anything positive for young
Humlum and Vestergaard (2025)	Denmark	Admin register + survey	DiD	Adopting against non-adopting workplaces and workers	Survey respondents linked to registers	Yes (22–25 share)	No change in the employment share at 22–25, effects above a third of a percentage point ruled out; null on earnings and hours; tasks reorganised
Bonin et al. (2026)	Denmark	IT survey + monthly employer-employee register	ES against own trend	Adopters against non-adopters, within industry-month	Surveyed private non-financial firms with at least 10 full-time employees	Yes	Adopters 11% below own trend by late 2025, under-30s 20%; hiring margin; firms under 100 FTE only
Tucker (2026)	US	QWI, private-sector flows	ES on industry-state cells	Exposed against less exposed industry-state cells	Near-universe of private-sector flows, industry-state-age tabulations	Yes (22–24)	Early-career employment –12% after ten quarters; hiring margin; monetary shocks up to a quarter
Chandar and Klein Teeselink (2026)	41 countries	Revelio Labs profiles + postings	Matched IV ES	Junior share, adopting against matched non-adopting affiliates	Public professional profiles; affiliates of multinationals	Yes (seniority)	Junior share –1.9 pp; senior +6.7%, junior –2.5% (ns)
This paper	Sweden	AGI register + Platsbanken	Poisson DiD	Young against older colleagues within employers, exposed against less exposed employers	Every employer, public sector and part-time work included; 104,217 employers identify	Yes	Young-to-older ratio at 22–25 –3.9% more at exposed employers, 2024–25 against 2023; relative gains rise with age; falls on young women

III.6. Published aggregates: a check whose coverage is not ours

Table A31 checks the age pattern in numbers whose construction is not ours. Statistics Sweden publishes employment by occupation and age band as an official aggregate in which the agency codes each worker’s occupation, where our design never codes a worker under 31 or reads a register year after 2019. For each published age band, the table reports the change since 2022 in the log employment gap between the top exposure quartile and the rest.

The 16–24 band, the closest published counterpart to the 22–25 group, is the band whose gap falls most by 2024. The bands 40–44 and 45–49 are flat, and every other band rises, so the direction of the age gradient is present in data the agency coded itself. The check is coarse: annual, in five-year bands, reaching 2024 but not 2025, with a closest band that includes teenagers, and on a register that changes method at reference year 2022, a break the agency itself flags. The check corroborates the age pattern but does not estimate its size.

Table A31: Change since 2022 in the top-quartile employment gap, published aggregates

Age band	2023	2024
16–24	+0.017	−0.006
25–29	+0.070	+0.021
30–34	+0.044	+0.054
35–39	+0.023	+0.032
40–44	+0.000	−0.002
45–49	−0.002	−0.002
50–54	+0.027	+0.051
55–59	+0.026	+0.039
60–64	+0.026	+0.045

Notes: Statistics Sweden’s published occupational employment register. Entries are the change since 2022 in the log employment gap between the top DAIOE exposure quartile and the remaining quartiles, within age band. Positive entries mean the gap widened in favour of exposed occupations.

Part IV: Coverage of the occupation register

The usual register design for AI and occupations gives every worker the occupation the register records for them and compares workers in exposed and less exposed occupations. This part measures what the lag in the occupation register does to that design, the occupation-classified design. Truncating the register as the lag does turns small positive estimates for past years into large negative ones, through re-coding and sample exclusion. Occupation is recorded in an annual register that appears at the start of the year after next: the register for 2024, which describes employment in November 2024, was published in January 2026 (Statistics Sweden, 2026d). Our research extract was delivered in 2025, before that release, so its occupation codes end with the 2023 register, while its monthly payroll declarations run up to and including June 2025. In the most recent years almost every worker therefore carries a code observed in an earlier year. The paper’s own exposure score reads no register file after 2019 (Section IV.4).

IV.1. Match rates, vintages and exclusions

Among declared employer, person and month records for workers aged 22 to 25, the share without an assignable occupation code rises from 8.4 per cent in 2023 to 12.8 per cent in 2024 and 17.8 per cent in 2025 (Table A44, Section VI.1). The rates below refer to narrower populations, each named with its rate.

Table A32 asks a narrower question: among employed workers aged 22 to 69 whom the occupation register has recorded, how many receive a code under the rule of a design that classifies workers by their own code, and how old is that code? Panel A shows that almost all do, in every year except 2019, the first year of the panel, where employment spells are left-censored. This does not contradict the rates above, because the denominators differ. Panel A counts only workers found in the register, so a worker it has never recorded, such as a new entrant, is outside it, and its near-zero shares after 2019 say nothing about such workers. On the full payroll denominator of Table A44, 3.6 to 6.0 per cent of workers aged 22 to 69 have no code under the same rule, and at 22 to 25 the share rises from 8.4 per cent in 2023 to 17.8 per cent in 2025. The rule assigns each worker the code in that year’s register up to 2022 and, from 2023, the most recent code in the 2023, 2022 or 2021 register, a fallback we call the cascade. The paper’s own exposure score, by contrast, reads the 2019 register and falls back only to earlier years.

What changes is the age of the code. In Panel B, almost every worker carries a code recorded that same year in 2023, almost none does in 2024, when the register for that year had not been published, and by 2025 almost all codes are two years old. A worker who changed occupation in 2024 is still recorded in the one they held in 2023. Panel C shows whom this reaches first: pooling 2020 to 2023, incumbents and recent hires carry a current code in about 99 per cent of person-employer pairs, labour-market entrants in 93 per cent, and young workers are disproportionately entrants.

That is why the paper’s exposure score is built from incumbents aged 31 to 69 and never from the young: it places the classification on the workers the register covers best and asks nothing of the codes of the workers whose employment is measured. Because the employment counts use no worker’s own code, the estimates cannot be split by the age of a code or into incumbents and new matches on that basis. A split that uses no code is available: workers on the employer’s payroll in November 2022, the launch month, against all others. At 22 to 25, τ is -0.327 (0.018) for the first group and -0.074 (0.023) for the second, on 65,595 and 82,481 employers. Both groups are selected on their employment histories. The first group admits no new members after November 2022, and its members age out of the young band. Conditioning on presence in that month also selects particular employment histories, so its coefficient is not an effect on established employees. The second group is everyone outside that selection. It includes earlier entrants who remain, so it does not measure hiring. Neither decomposes the change into hiring and separations.

Table A32: Occupation-code coverage in the occupation-linked population, and the age of the code.

<i>Panel A. Share of employment excluded for want of any code, by age band</i>				
Year	22–25	26–30	41–49	50+
2019	1.28%	5.65%	3.94%	5.81%
2021	0.00%	0.00%	0.00%	0.00%
2023	0.00%	0.00%	0.00%	0.00%
2025	0.00%	0.00%	0.00%	0.00%
<i>Panel B. Age of the occupation code, share of all employment</i>				
Year	Current	One year old	Two years old	Older
2023	99.7%	0.2%	0.0%	0.0%
2024	0.0%	99.9%	0.1%	0.0%
2025	0.0%	0.0%	99.9%	0.1%
<i>Panel C. Match rate by worker group, 2020–2023 pooled</i>				
Group	Share carrying a code			
Incumbent	99.1%			
Recent hire	99.4%			
Entrant, first panel year	92.9%			

Panel A: the share of employed workers aged 22 to 69 with a birth year in the register used for whom no occupation code can be assigned, taking each year’s own register up to 2022 and, from 2023, the most recent code in the 2023, 2022 and 2021 registers; it is non-zero only in 2019, where employment spells are left-censored. Workers with no record in the register used are outside the denominator. Panel B: the age of the code carried, by year; the register is published with a two-year lag. Panel C: the share of person-employer pairs carrying a current code, by worker group, 2020 to 2023 pooled.

IV.2. Event studies by code vintage and by margin

Table A33 splits the event study of the occupation-classified design at ages 22–25 by the register year that supplied each worker’s code, the obvious remedy for stale codes. The design is estimated by Poisson pseudo-maximum likelihood. The three columns disagree by an order of magnitude: at the 2025 endpoint one returns -1.32 , a fall of nearly three quarters on a subsample selected as described below. The subsamples collapse in the later periods, because a worker can carry a 2021 code in 2025 only if no later register observed them, which is a statement about tenure and mobility rather than occupation.

Conditioning on the vintage of a code is conditioning on how recently the register saw the worker, which depends on the entry, exit and job changes the design is meant to measure, so the sample is selected on the outcome and a subsample of fresh codes does not isolate the measurement problem. Section IV.3 measures it instead by holding the sample as fixed as the register allows and varying what the classifier may know.

Table A33: The half-year event study estimated separately on workers coded from each register vintage, ages 22–25.

Period	Coded from 2021	Coded from 2022	Coded from 2023
2019H1	−0.010 (0.016)	−0.010 (0.016)	−0.005 (0.015)
2019H2	−0.037 (0.017)	−0.037 (0.017)	−0.031 (0.016)
2020H1	−0.041 (0.015)	−0.041 (0.015)	−0.035 (0.015)
2020H2	−0.053 (0.016)	−0.052 (0.016)	−0.047 (0.015)
2021H1	−0.021 (0.011)	−0.021 (0.011)	−0.018 (0.011)
2021H2	−0.054 (0.012)	−0.054 (0.012)	−0.051 (0.011)
2022H1	+0.000 (0.000)	+0.000 (0.000)	+0.000 (0.000)
2022H2	−0.047 (0.009)	−0.047 (0.009)	−0.050 (0.008)
2023H1	−0.294 (0.341)	−0.201 (0.120)	+0.006 (0.010)
2023H2	−0.557 (0.405)	−0.024 (0.138)	−0.018 (0.012)
2024H1	−0.344 (0.463)	−0.358 (0.278)	−0.283 (0.015)
2024H2	−0.539 (0.373)	−0.643 (0.268)	−0.347 (0.017)
2025H1	−0.532 (0.533)	−1.317 (0.354)	−0.587 (0.026)
<i>Person-months of workers aged 22–25 in each column</i>			
2023H1	629	7,837	2,173,825
2023H2	690	3,681	2,309,084
2024H1	488	1,082	2,119,020
2024H2	1,007	1,868	2,245,754
2025H1	925	1,993	2,095,368

Each column restricts the occupation-classified event study, re-estimated by Poisson pseudo-maximum likelihood, to workers whose code comes from the stated register year; reference 2022H1, standard errors clustered by employer. Conditioning on code vintage conditions on how recently the register observed a worker, which depends on tenure, entry and mobility, so the subsamples shrink sharply in the later periods and no column estimates an effect of exposure. The lower panel gives the person-months behind each column from 2023, when the columns first differ: the 2021 and 2022 columns rest on between 500 and 8,000 person-months a half-year against about 2.1 to 2.3 million in the 2023 column.

IV.3. Re-estimating on the codes available at the time

Table A34 and Figure A7 re-estimate the occupation-classified design on the codes available at the time. For 2019 to 2023 the register is now complete, so each worker’s final recorded occupation is known, and we call these the completed codes. The design is estimated twice on those years with the same specification. The first estimate uses the completed codes. The second uses the codes as they would have stood had the register stopped in 2021, or in 2022. The difference between the two estimates is the effect of the publication lag. The truncated version loses the workers the truncated register cannot code, since that loss is part of what the lag does, so the gap combines re-coding with a change in who is included.

Table A35 separates the two on the employers common to all three arms. Holding the completed codes fixed and changing only who is included moves the coefficient by -0.011 at the 2021 truncation and -0.009 at 2022. Changing only the codes, on the same workers, moves it by -0.276 and -0.139 . Re-coding accounts for about 96 and 94 per cent of the difference and sample inclusion for the rest, a smaller but non-zero part. The arms share their employers, and the observations each fit retains after the Poisson drops of all-zero groups differ by under two per cent of the input cells across them (Table A35). The dates

mirror the real panel relative to the register. For a register stopped in year T , the pseudo rate rise falls in April of $T - 1$ and the pseudo launch in December of $T - 1$, just as the real rise and launch fall in 2022 and the last real register year is 2023. The post-launch window therefore runs into the two years after T , whose codes are stale, as the real one runs into 2024 and 2025. These are historical decompositions. They do not show that missing workers are unimportant in every period, or that the same shares apply to that design’s estimate for 2024 and 2025.

With the completed codes the design finds nothing: the coefficient is small, positive and indistinguishable from zero. With the codes available at the time, in the same years but without the workers the truncated register cannot code, it finds a large and precise decline among the young, which combines re-coding with sample inclusion. Truncating at 2022, the milder case, halves the artefact without removing it. On the real years the same design returns a coefficient of about -0.17 at 22–25, so the lag alone can produce more than the whole coefficient. On these data such a design does not separate a real effect from the publication lag, which is why the paper does not use it.

Table A34 re-estimates the coefficient of about -0.17 that the occupation-classified design returns on 2021–2025. That coefficient, the submitted paper’s pooled OLS estimate of -0.010 and its endpoint event-study estimate of -0.055 are estimates of different quantities, and none is a version of τ .

The problem is not particular to Sweden: any design that classifies workers by their own recently recorded occupation, in a register published with a lag, is exposed to it, and the size of the exposure can be measured on the researcher’s own data by this procedure.

The paper’s own design classifies employers once, in 2019, from the occupations of incumbents aged 31 and over, so truncating the register at any later year leaves its exposure, outcome and sample unchanged. Statistics Sweden dates one in fifty of the 2019 codes to its 2020 collection, all recorded before the launch of ChatGPT. Scoring each employer from the codes dated 2019 alone moves the estimate at 22–25 by 0.005 log points, under a tenth of its size, and 97.8 per cent of employers keep their quartile (Section IV.4).

Table A34: Codes available at the time: what the register’s lag alone produces in years for which the completed codes are known.

Register truncated at	Completed codes	Codes available at the time	Difference
2021	+0.0193 (0.0129)	−0.2875 (0.0171)	− 0.3068
2022	+0.0176 (0.0111)	−0.1452 (0.0122)	− 0.1627

The coefficient is the occupation-classified design’s $\hat{\gamma}_2$ on 2019–2023, years in which the completed register is available. The second column re-estimates it after truncating the register at the stated year and rebuilding the fallback to earlier registers as it would then have been, which imposes on those years the staleness that 2024–25 inherit. That version also loses the workers the truncated register cannot code, so the difference combines re-coding and sample inclusion. The difference is the gap between the two columns. At the 2021 truncation the lag moves an estimate of $+0.019$, indistinguishable from zero, to -0.288 , a difference of -0.307 . On the real years the design returns about -0.17 , so the lag alone can produce more than the whole of it. The design the paper reports reads no occupation file after the 2019 vintage, and its employment counts need no current code. Restricting its score to codes whose recorded observation year is 2019 moves its estimate at 22–25 by $+0.005$, and 97.8 per cent of employers keep their quartile.

Table A35: Codes available at the time, decomposed: sample inclusion against re-coding, on common employers

Register truncated at	Version			Difference			Employers
	A	B	C	B-A	C-B	C-A	
2021	-0.0020 (0.0131)	-0.0134 (0.0127)	-0.2897 (0.0172)	-0.0115 (0.0026)	-0.2762 (0.0129)	-0.2877 (0.0136)	35,068
2022	+0.0047 (0.0111)	-0.0042 (0.0110)	-0.1436 (0.0122)	-0.0090 (0.0013)	-0.1394 (0.0064)	-0.1484 (0.0068)	38,599

The occupation-classified design’s post-launch coefficient at 22–25 (employer $\times$ exposure-quartile $\times$ month cells, 2019–2023; rate-rise and launch interactions with the top quartile; employer-by-quartile and employer-by-month effects; Poisson; standard errors clustered by employer). Pseudo-dates for truncation year T : rate rise April $T-1$, launch December $T-1$, reference period before the rate rise. A: completed codes, all workers. B: completed codes, the workers the truncated register can code. C: codes available at the time (each year’s own register to T , then the files of T , $T-1$ and $T-2$), the same workers as B. All three versions are estimated on the employers common to them, and the differences come from one stacked fit, so B-A isolates sample inclusion and C-B re-coding. The panel is balanced and zero-filled, and the Poisson fit drops the cells of all-zero employer-by-quartile groups and the singletons they leave, about a third of the input: at the 2021 truncation the three versions retain 4.55, 4.47 and 4.45 million of 6.8 million cells, at 2022 4.87, 4.83 and 4.82 million of 7.4 million, and the stacked fit retains exactly the sum of the three.

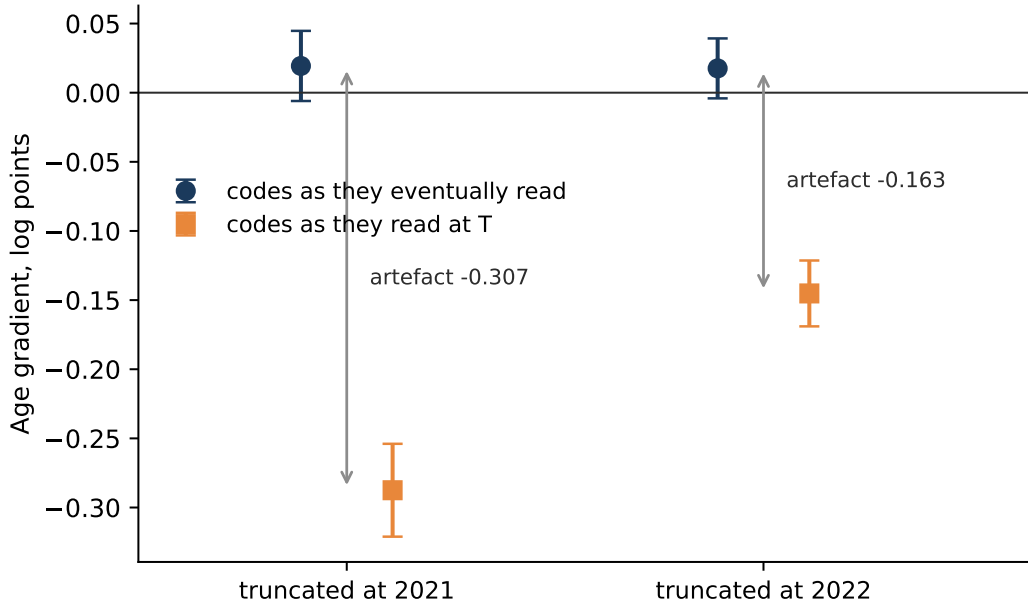


Figure A7: Codes available at the time: for each truncation of the occupation register, the occupation-classified design’s coefficient with the completed codes and with the codes the cascade would have produced had the register stopped that year. The specification is identical within each pair; the truncated version carries older codes and loses the workers it cannot code.

IV.4. The codes the exposure score is built from

Table A36 reports which occupation codes the exposure score reads. Each incumbent aged 31 to 69 on an employer’s 2019 payroll is given the occupation the register records

for them in 2019. Where the 2019 file holds none, the score falls back to 2018, then 2017, 2016 and 2015, and no further; it never looks forward. The 2019 file alone codes over nine incumbents in ten, and the carry-back raises that by a further two points, so one resolved code in forty comes from a year before 2019. Coverage is lowest among incumbents aged 50 and over, and almost every resolved code carries four digits.

The register also records the year in which each occupation was observed, which is the sharper measure of how fresh the codes are. Seven incumbents in ten were observed in 2019 itself and a further one in twelve in 2018. One in fifty was observed in 2020 (1.94 per cent), as Statistics Sweden completed part of its 2019 file from the following year's collection. Those codes were recorded before the launch of ChatGPT and two years before the later period begins. The paper's claim is therefore that no register *file* after 2019 is read, not that no code was observed after 2019: the score reads the 2019 register file and, where it holds no code, the files back to 2015. A construction that also read the 2020 and 2021 files for incumbents still uncoded would add 71,882 codes and leave the quartile of every one of the 262,089 scored employers unchanged.

Every incumbent is scored with a three-digit lookup table: the 2019 national employment-weighted mean of the four-digit scores within each three-digit group. This holds even though almost every code carries four digits. The reason is bias. Workers lacking a four-digit code are not a random subset, so a rule that used four digits where available would give a precise score to employers whose coding is complete and a smoothed one to the rest, and quartile assignment would then depend partly on coding completeness. Applying the three-digit table to every incumbent makes the smoothing common to every employer, and it costs little. Almost all of the variance in the four-digit score lies between three-digit groups, and the estimate at 22–25 moves by less than 0.001 across the three scoring rules in the last block of Table [A36](#).

Part V: The within-employer design on public advertisements

Public advertisements permit a separate within-employer comparison, of exposed against less exposed occupations inside each employer's own advertisements, that uses no occupation register; this part reports it.

Table A36: What the exposure score is built from: the 2019 occupation codes of the employer’s own incumbents

	Incumbents	Coded from 2019 alone (%)	Coded, with carry-back (%)	Scored (%)
31–34	474,587	91.6	94.7	94.3
35–40	672,986	92.6	95.3	95.1
41–49	1,063,594	93.7	96.0	95.8
50 and over	1,907,832	90.9	93.1	92.9
All incumbents, 31–69	4,118,999	92.0	94.4	94.2
<i>Employers, and the floor</i>				
With an incumbent aged 31 to 69	294,887			
scored at a floor of one person-month	271,047			
scored at a floor of three person-months	267,605			
scored at a floor of five person-months (reported)	262,089			
<i>What the scoring level changes</i>				
Employers moving quartile against four-digits-where-usable (%)				9.7
rank correlation of the two employer scores				+0.977
Employers moving quartile against four-digits-only (%)				9.8
rank correlation of the two employer scores				+0.976

Incumbents are the employees aged 31 to 69 on the employer’s 2019 payroll, the workers the score averages over. A code is taken from the 2019 register and otherwise from 2018 back to 2015, the carry-back column; 2.5 per cent of resolved codes come from those earlier years, and none from a later file. The average employer draws 7.4 per cent of its codes from them, because small employers depend more on the older files. Scored is the share the DAIOE index can price. Every incumbent is scored from the three-digit book, and the last block reports what scoring at four digits changes.

V.1. Design and sample

From January 2021 about 99 per cent of Platsbanken advertisements carry the employer’s organisation number, so a within-employer comparison can be estimated on public data: not Part III’s comparison of ages, but one of exposed against less exposed occupations inside the same employer’s advertisements. The employer and the occupation are recorded at publication, so no occupation register, and no register lag, enters. We count distinct advertisements by employer, exposure quartile and month from January 2021 to June 2026 and estimate Equation (1) of the paper with employer $\times$ quartile and employer $\times$ month effects, by Poisson, clustering by employer. Employers with fewer than five advertisements over the window are dropped, and identification requires an employer to post in both an exposed and a less exposed occupation, which 12,141 employers do.

An entry-level variant applies the same floor and eligibility rule to entry-level advertisements alone, which leaves 1,265 employers. An advertisement counts as entry-level when its text contains one of a short list of Swedish terms by which employers signal that no experience is required: *nyexaminerad* or *nyutexaminerad* (newly graduated), *junior*, *trainee* or *traineeprogram*, *ingen erfarenhet* (no experience) and *utan (tidigare) erfarenhet* (without previous experience). The list was fixed before this exercise and applied unchanged to the whole archive.³ It signals how a post is advertised, not who is hired, and 2.8 per cent of distinct advertisements carry it.

³The advertisement field that records experience requirements directly is present too rarely to be usable.

The window before the rate rise is fifteen months, since organisation numbers begin in January 2021, the same baseline before the rate rise as the register design. The counting unit is the distinct advertisement, not the vacancy or the hire, so the design speaks to posting behaviour, not to employment.

V.2. Results

Table A38 reports the design on all identifying employers, excluding staffing agencies and public employers, and on entry-level advertisements alone. The rate-rise interaction is indistinguishable from zero in every variant. After ChatGPT an employer’s advertisements in top-quartile occupations fall relative to its other advertisements: the coefficient is -0.158 on all employers, about 15 per cent, and -0.180 once staffing agencies and public employers are removed, so the decline is not an artefact of intermediated hiring.

Table A37 checks two things. Applying the five-advertisement floor and the identification screen to January 2021 to March 2022 alone, and keeping every later cell including zeros, gives -0.156 on the employers that qualify before the rate rise, so the whole-window eligibility rules do not select the result. On entry-level advertisements the estimate is -0.196 , against -0.160 for all advertisements of the same employers, and the entry-level differential estimated in one panel is -0.044 (SE 0.066), so the entry-level response is not distinguishable from the employer’s other advertising.

Table A37: Within-employer posting design: eligibility before the rate rise, and the entry-level differential on a common employer sample

Sample	Post-rate-rise $\times$ High	Post-launch $\times$ High	Employers	<i>N</i>
Eligibility before the rate rise, every quartile cell	-0.003 (0.011)	-0.156^{***} (0.013)	5,277	624,079
Eligibility before the rate rise, quartile cells before it only	0.008 (0.011)	-0.155^{***} (0.013)	5,277	569,624
As the first row, excluding staffing agencies and public employers	-0.020 (0.022)	-0.180^{***} (0.022)	4,230	414,624
All advertisements, employers passing both screens	-0.005 (0.014)	-0.160^{***} (0.016)	1,265	228,317
Entry-level advertisements, the same employers	-0.007 (0.117)	-0.196^{***} (0.065)	1,265	49,246
<i>Entry-level differential, one panel (employer $\times$ quartile $\times$ entry flag $\times$ month)</i>				
Post-rate-rise $\times$ High, non-entry advertisements	-0.008 (0.014)		1,265	273,809
Post-launch $\times$ High, non-entry advertisements	-0.152^{***} (0.016)		1,265	273,809
Post-rate-rise $\times$ High $\times$ Entry	0.001 (0.120)		1,265	273,809
Post-launch $\times$ High $\times$ Entry	-0.044 (0.066)		1,265	273,809

Poisson, employer $\times$ quartile and employer $\times$ month effects, clustered by employer, January 2021 to June 2026; the one-panel differential adds the entry flag to both effects. Eligibility before the rate rise applies the five-advertisement floor and the two-quartile eligibility rule to January 2021 to March 2022 only. Entry-level as defined in Section V.1; the flag fires on 2.4 to 3.4 per cent of advertisements by year and on 0.9, 2.2, 3.2 and 4.0 per cent in exposure quartiles 1 to 4. The Poisson fits drop employer-by-quartile groups that are zero throughout. *** $p < 0.01$.

Table A39 reports the post-ChatGPT interaction across industry sections, employer age and employer size. It is negative in all thirteen cells and significant at five per cent in eleven; information and communication is the weakest cell and not significant, and the public sector is close to the average. Within the employer, advertising shifts away from exposed occupations broadly across sectors and sizes, even though market-wide postings show no significant additional decline. Because the entry-level response does not differ from the rest, the paper reads this as evidence on how employers reallocate their advertising across occupations, not as a second estimate for young workers.

Table A38: Within-employer posting design, public advertisements 2021–2026

Sample	Post-rate-rise $\times$ High	Post-launch $\times$ High	Employers	<i>N</i>
All identifying employers	0.004 (0.011)	−0.158*** (0.013)	12,141	881,443
Excluding staffing agencies and public employers	0.001 (0.021)	−0.180*** (0.020)	10,509	638,248
Entry-level advertisements only	−0.007 (0.117)	−0.196*** (0.065)	1,265	49,246

Notes: Poisson, employer $\times$ quartile and employer $\times$ month fixed effects, standard errors clustered by employer. The unit is the distinct advertisement. The population is employers whose advertisements carry an organisation number and who post in both exposed and less exposed occupations in the window. Staffing agencies are identified by industry code SNI 78 and public employers by the organisation-number prefix. ***, **, * denote significance at 1, 5 and 10 per cent.

Table A39: Within-employer posting design, by employer characteristics

Dimension	Cell	Post-launch $\times$ High	Employers
Employer age	under 10 years	−0.187*** (0.047)	3,777
Employer age	10 years or more	−0.156*** (0.013)	7,745
Industry	other private	−0.244*** (0.030)	4,232
Industry	C manufacturing	−0.203*** (0.044)	1,204
Industry	K finance	−0.181 (0.123)	171
Industry	N administrative and support	−0.180*** (0.033)	1,369
Industry	M professional services	−0.161*** (0.040)	1,333
Industry	public	−0.132*** (0.016)	780
Industry	G trade	−0.116** (0.055)	1,489
Industry	J information and communication	−0.101 (0.089)	955
Employer size	medium	−0.204*** (0.059)	1,874
Employer size	large	−0.157*** (0.013)	6,128
Employer size	small	−0.150** (0.074)	1,112

Notes: As Table A38, estimated separately within each cell.

Part VI: Data documentation

This part documents each data source, its coverage, how the registers are joined, and how the data can be accessed.

VI.1. Data sources and access

Table A40 summarises the data sources; documentation for each follows.

VI.1.1. Platsbanken

Job advertisement data from the Swedish Public Employment Service (Arbetsförmedlingen, AF), published under CC0 licence. Each record contains: ad identifier, publication date, SSYK 2012 four-digit occupation code, number of vacancies, municipality, employer identifier, and source type. For 2020–2026 we use the historical bulk archive (<https://data.jobtechdev.se/annonser/historiska/>), which publishes closed quarters and covers the first half of 2026; no month is taken from the JobStream API (<https://jobstream.api.jobtechdev.se>), whose snapshot endpoint returns only the advertisements live at extraction. The yearly and closed-quarter files are deduplicated on the advertisement identifier before aggregation. In recent years about 700,000 advertisements have been published on Platsbanken each year.

Table A40: Data sources.

Source		Period	Frequency	Unit	Key variables
Platsbanken (AF)		2020–2026	Monthly	Job ad	SSYK 4-digit, vacancies, municipality, employer
AGI register (SCB/MONA)		2019–2025	Monthly	Worker-employer	Employment, compensation, employer ID, person ID
LISA (SCB/MONA)	Individ	2015–2023	Annual	Individual	SSYK 4-digit, birth year, sex
Education register (SCB/MONA)		2023	Annual	Individual	SUN 2020 field and level
DAIOE		2023	Cross-section	SSYK 4-digit	GenAI exposure percentile rank
OMXS30		2020–2026	Daily monthly	→ Index	Closing price
Riksbank policy rate		2020–2026	Event dates	—	Policy rate level

Coverage and representativeness. Platsbanken is the largest recruitment site in Sweden, and job postings are increasingly used in labour market research (e.g. [Hansen et al., 2023](#); [Liu et al., 2026](#); [Lambert and Schindler, 2026](#)). Four facts bear on its coverage. Most vacancies in Sweden are filled through personal contacts and direct approaches, but the share filled via AF has been stable or slightly rising: in 2006, 9% of already-employed job changers and 13% of unemployed individuals who found work did so via AF, and by 2023 these shares had risen by 1 and 2 percentage points. Private and local-government employers were required by law (SFS 1976:157) to post vacancies at AF until the obligation was repealed in 2007 (SFS 2007:396), and only central government retains its own requirement (SFS 1984:819). [Cronert \(2022\)](#) finds that the repeal lowered the rate at which employers notified vacancies to the Public Employment Service by between 7.5 and 20.2 per cent of its mean for the labour market as a whole. The repeal came thirteen years before our window, so it bears on the level of coverage, which the occupation effects absorb, and not on its change over 2020 to 2026. Vacancies are also posted on LinkedIn, Monster and sector-specific sites, so Platsbanken may undercount some industries and occupations, although posting there is free and AF redistributes postings to other platforms. And employers may redeploy existing staff to AI-related tasks, so postings can understate adoption.

A fixed difference in coverage between exposure groups is absorbed by the occupation effects; the threat is a change in relative coverage. If exposed occupations moved to other platforms faster after 2022, Platsbanken would understate their advertising increasingly over time, biasing the posting interactions downward, and the same drift inside an employer could mimic the within-employer decline of Part V. That drift cannot be measured here, so it is a limitation of the posting margin. It does not reach the employment margin, which uses no advertisements.

VI.1.2. DAIOE

The Dynamic AI Occupational Exposure index maps AI benchmark performance to occupational task content. The GAI variant focuses on capabilities relevant to large language models and image generation. Publicly available; see [Engberg et al. \(2024\)](#).

VI.1.3. OMXS30

The stock-market series are the OMX Stockholm 30 (ticker `OMX`) and the OMX Stockholm All-Share, daily closing prices from Yahoo Finance averaged by month and indexed to 100 at the February 2020 mean. Both run to August 2026.

VI.1.4. Riksbank policy rate

Manually compiled from Riksbank press releases. Key dates verified against <https://riksbank.se>.

VI.1.5. Administrative employment data (SCB/MONA)

The employment analysis draws on the ORU-MICRO-AI employer-employee database of Örebro University, which links administrative registers from Statistics Sweden at the person level. Three supply variables here: AGI (*arbetsgivardeklaration*, the monthly employer declarations to the Swedish Tax Agency) gives the encrypted person identifier, the employer identifier and monthly employment status; the annual LISA register, with its embedded occupation register (*Yrkesregistret*), gives the SSYK 2012 four-digit occupation code, birth year and sex; and the education register (*Utbildningsregistret*) gives each worker's highest completed education, classified by SUN 2020, from which the two-digit field is taken.

The education register enters only through the cuts by field in Section III.2, where it divides the sample; no estimate is scored from education. The record is the worker's own most recent completed education. Because the register is annual and published with a lag, the 2023 record is carried forward to 2024 and 2025, so a young worker who completes a further degree during the window keeps the field recorded in 2023, and the fields are kept broad.

Source and coverage. The AGI register originates from mandatory monthly employer declarations to the Swedish Tax Agency (*Skatteverket*), introduced in January 2019: every employer making PAYE (*A-skatt*) salary payments reports each individual payment, so the register covers the full universe of employer-employee matches, part-time and temporary workers included, with no sampling error. The unit is the employer-employee relationship: a person-month is counted when the employer files an individual declaration record for the person in that month, whatever payment it reports, so a person is counted once per employer and month and cash pay is not required. Self-employed individuals operating under F-skatt (sole traders, platform workers) do not appear. A shift towards such arrangements among young workers would reduce measured employment, which the month-by-age effects absorb where it is common to employers, and which would bias the estimate only if it ran differently at more and less exposed employers.

Table A41 reports what that rule admits. Of the 284.7 million counted person-months, 94.0 per cent carry cash pay subject to employer contributions: 96.5 per cent at ages 22 to 25 and 93.4 per cent at 31 to 69, where two thirds of the remainder carry an occupational pension paid through the employer's declaration. At 22 to 25 the remainder carries neither a pension nor, in the main, a taxable benefit. On the headline contrast the cash-pay share moved by -0.8 percentage points at 22 to 25 and -0.6 at 31 to 69 (top quartile minus the rest, later period minus interim), so, computed from these shares, counting only cash-pay person-months would move $\hat{\tau}$ by -0.0025 , a quarter of its standard error, towards a larger

decline. Of the 2019 person-months aged 31 to 69 behind the employer score, 94.9 per cent at top-quartile employers and 92.5 per cent at the others carry cash pay. A record carrying only a pension counts towards the floor. It also enters the score when the person holds a code in the 2019 register or in the earlier years the score falls back to.

Table A41: What the counting rule admits: cash pay behind the counted person-months

<i>Panel A. Counted person-months with cash pay subject to employer contributions, per cent</i>					
	Jan 2021–Mar 2022	Apr–Nov 2022	Dec 2022–Dec 2023	Jan 2024–Jun 2025	
22–25, top quartile	94.8	96.6	95.5	93.8	
22–25, other quartiles	97.1	97.7	97.3	96.4	
26–30, top quartile	96.8	97.4	97.2	96.7	
26–30, other quartiles	97.5	98.0	97.8	97.2	
31–69, top quartile	95.6	95.8	94.1	93.7	
31–69, other quartiles	92.8	93.2	92.8	93.0	
<i>Panel B. Top quartile minus other quartiles, percentage points</i>					
	Jan 2021–Mar 2022	Apr–Nov 2022	Dec 2022–Dec 2023	Jan 2024–Jun 2025	Later minus interim
22–25	–2.29	–1.06	–1.80	–2.60	–0.80
31–69	+2.85	+2.58	+1.27	+0.70	–0.57
<i>Panel C. Counted person-months without cash pay, all employers and periods</i>					
	Per cent of counted	With a pension amount, per cent	With a taxable benefit, per cent		
22–25	3.49	0.2	3.8		
26–30	2.80	0.3	9.0		
31–69	6.62	65.5	6.8		
<i>Panel D. The 2019 scoring population: person-months aged 31–69 behind the employer score</i>					
	With cash pay, per cent	Of the remainder: pension, per cent	benefit, per cent	Person-months	
Top quartile	94.9	35.6	19.4	12,165,931	
Other quartiles	92.5	76.9	1.5	36,497,316	
All scored employers	93.0	67.0	4.8	49,596,472	

A counted person-month is any distinct person for whom the employer files an individual declaration record in the month, with a birth year in the 2023, 2021 or 2019 individual register and an age of 22 to 69; cash pay is not required. Cash pay: the declaration's cash remuneration subject to employer contributions exceeds zero; pension: an occupational-pension amount is declared; benefit: a car or other taxable benefit is declared. Exposure groups by the paper's 2019 employer score; employers the score does not reach are in the row totals of Panel C only. On these shares, counting cash-pay person-months alone would move $\hat{\tau}$ at 22–25 by the later-minus-interim, top-minus-rest change in the log share of the young band less that of the older bands, -0.0025 , about a quarter of its standard error and towards a larger decline; this is a calculation on the shares, not a refit.

The employment design scores an employer whose incumbents aged 31 to 69 account for at least five person-months on its 2019 payroll and at least one of whom holds a 2019 occupation the index scores. The occupation mix is weighted by the November 2019 head count; the floor and the national quartile cut are taken on incumbent person-months. The employer must also hold the young band and at least one older band in the panel window. Because the score book is fixed in 2019, the share of person-months at an employer it scores falls from 97.7 per cent in 2019 to 91.7 per cent in 2025, as employers founded after 2019 take on staff.

Occupation coding, and how SSYK relates to ISCO and SOC. Occupations are recorded in SSYK 2012, the Swedish standard classification, employer-reported, held in a separate annual occupation register (*Yrkesregistret*) and published with a lag of about two years. SSYK 2012 is the Swedish implementation of ISCO-08: the two share their structure down to the three-digit group and SSYK adds Swedish detail at the fourth digit, so the crosswalk between them is defined at the four-digit level but is not the identity, and because the two use overlapping numeric codes with different meanings they are never merged on the raw code. Statistics Sweden's own crosswalk is used throughout.

This matters because the exposure index is not built on Swedish data: DAIOE is estimated on the abilities O*NET records for US occupations, coded in the US Standard Occupational Classification, and reaches Swedish occupations through the chain SOC 2018 to SOC 2010 (the US Bureau of Labor Statistics crosswalk), SOC 2010 to ISCO-08, and

ISCO-08 to SSYK 2012 (Statistics Sweden’s crosswalk). The [Eloundou et al. \(2024\)](#) score and the [Dingel and Neiman \(2020\)](#) teleworkability classification reach SSYK by the same route. Every step of that chain is many-to-many, one reason the measure is read at quartiles rather than as a fine gradient (Section II.5).

The design the paper reports reads no occupation register file after 2019: the register enters only through the 2019 occupations of each employer’s incumbents aged 31 to 69, which give the employer its exposure, and through the diagnostics of Part IV, which measure what classifying every worker by their own most recent code would do.

Stability and comparability. The AGI source has been stable since January 2019, with no change during 2019–2025 to the reporting of the employer, the person or the month, and the SSYK 2012 classification has been unchanged throughout. The annual employment registers (RAMS/BAS), by contrast, changed method at reference year 2022, which falls within the dates the design compares. Using AGI for monthly employment and Yrkesregistret for occupation avoids that break.

The counts rebuilt from the raw declarations. The employer, age band, sex and month counts were rebuilt by a separate query from the raw declarations, reading only distinct employer, person and month triples and the birth year and sex of the 2023, 2021 and 2019 individual registers, with no occupation field of any year and no intermediate file. Every cell of every year from 2021 to 2025 equals the count the paper uses, split by sex and pooled, with no cell present on only one side, and τ on the rebuilt counts is -0.039862 (0.010245) against -0.039862 (0.010245) on the paper’s. Pooled counts are distinct employer, person and month records rather than sums over sex. The 2019, 2021 and 2023 individual registers each hold 8.5 to 8.7 million persons. Of these, 8, 7 and 9 persons respectively are recorded under two sex codes, and a sum over sex counts them twice. A sex-summed count differs from the pooled one in 657 of the 24,884,226 panel cells, every one of them such a person. Removing those persons leaves the female differential at -0.0714 (0.0109) to four decimals.

Non-match rates and sample attrition. The employment counts do not require a current occupation code, so a missing or stale current code does not determine which workers are counted. Baseline scoring, demographic linkage and the sample restrictions are separate questions. A worker enters with a birth year, a sex and a payslip. Birth year and sex are read from the 2023, 2021 and 2019 individual registers. A worker in none of the three cannot be linked and is not counted, and such workers include those first registered in Sweden after 2023. Table [A42](#) counts those payslips. They are about 0.2 per cent of employer-declaration person-months, and the share does not rise once the register ends: it averages 0.279 per cent over 2019 to 2023 against 0.203 over 2024 and 2025, because birth year and sex, unlike occupation, can be read from any of the three vintages and almost every worker appears in one.

The share is higher in the top exposure quartile than in the lower three, by 0.051 percentage points on average over 2020 to 2022 and by 0.042 over 2024 and 2025. A constant gap is absorbed, since the design compares changes, and the gap narrows by 0.009 percentage points once the register ends. If the unlinked were young in the same proportion in both periods, the young workers of exposed employers are therefore, if anything, better counted in the later period. Table [A43](#) does the calculation on the headline sample and contrast.

At top-quartile employers the unlinked share is 0.262 per cent of declared person-months in the interim period and 0.240 in the later one, against 0.196 and 0.200 at other employers, so the gap between the groups narrows by 0.027 percentage points once the register ends. This again works against a spurious decline if the unlinked were young in the same proportion in both periods, and the allocations below assess the age-specific contrast directly.

For τ to be zero, the band aged 22–25 at top-quartile employers would have to be 4.1 per cent larger in the later period than counted, 41,303 person-months. That is three quarters of every unlinked person-month at those employers in that period, of any age, whereas the band is 4.4 per cent of declared employment. Adding every unlinked person-month at top-quartile employers to the band, in the interim and later periods as they occur, gives $\tau = -0.041$ (SE 0.010). Adding only the later-period excess over each employer’s own interim rate gives -0.031 (0.010). For the female differential the tipping point is 7.4 per cent of young women’s later-period counts, two thirds of the unlinked total where young women are 2.1 per cent of employment, and the two allocations give -0.072 (0.011) and -0.056 (0.011). The declaration carries no age of its own, so the calculation assigns every unlinked person-month to the young. It is a counterfactual on the counts, not a bound.

Table A42: The paylips the panel never counts: the share of employer-declaration person-months whose worker reaches none of the 2023, 2021 and 2019 individual registers.

Year	All employers	Top exposure quartile	Quartiles 1 to 3
2019	0.612	0.595	0.616
2020	0.205	0.263	0.187
2021	0.187	0.218	0.179
2022	0.188	0.216	0.179
2023	0.204	0.258	0.192
2024	0.195	0.231	0.192
2025	0.210	0.248	0.204

Per cent of employer-declaration person-months. A worker enters the employment panel only if a birth year and a sex can be read from the 2023, 2021 or 2019 individual register; the table reports the complement. Quartiles are the employer’s 2019 occupation-mix exposure quartile, and person-months at employers the 2019 score book does not score are outside the two quartile columns but inside the first. 2019 is the first year of the employer declarations and its gap takes the opposite sign, so the pre-period averages quoted in the text run from 2020; 2025 covers the first half of the year. The employer declaration carries no age of its own, so the share cannot be broken down by age band.

Non-match rates in a design the paper does not use. The rates below concern an occupation-classified design, one that classifies each worker by their own occupation code, as the submitted version of the paper did. The paper’s own employment counts use no worker’s code and are unaffected. The non-match rate shows which workers such a design can see in these years, and Part IV shows what it does to the design: on past years, codes as they stood at the time produce a large decline among the young that the completed codes do not (Section IV.3). Three quantities are easily confused.

The first is the *overall* non-match rate: the share of all employer-declaration person-months for which the occupation register holds no code. Table A44 rebuilds it from the raw declarations. The denominator is every distinct employer, person and month in the

Table A43: The unlinked paylips on the headline sample: their share by exposure group and period, the tipping point, and τ under two illustrative allocations

<i>Panel A. Unlinked paylips, per cent of declared person-months, headline employers</i>				
	Jan 2021–Mar 2022	Apr–Nov 2022	Dec 2022–Dec 2023	Jan 2024–Jun 2025
Top exposure quartile	0.222	0.228	0.262	0.240
Other employers	0.182	0.183	0.196	0.200
Difference, percentage points	0.040	0.045	0.066	0.040
Unlinked person-months, top quartile	38,988	22,229	44,186	55,634
<i>Panel B. What the unlinked person-months could do to τ</i>				
	Ages 22–25		Young women minus young men	
τ as estimated	−0.0399 (0.0102)		−0.0714 (0.0109)	
Proportional increase in the young band's later-period counts at top-quartile employers that sets τ to zero	4.1 per cent		7.4 per cent	
in person-months	41,303		36,476	
as a share of all unlinked person-months at those employers in the later period, any age	0.74		0.66	
The band's own share of declared person-months at those employers	4.4 per cent		2.1 per cent	
τ with every unlinked person-month at top-quartile employers added to the band	−0.0410 (0.0100)		−0.0724 (0.0113)	
person-months added	99,820		96,244	
τ with only the later-period excess over each employer's interim rate added	−0.0311 (0.0102)		−0.0564 (0.0111)	
person-months added	7,889		7,147	

Panel A: the 104,217 employers of the headline panel at 22–25; declared person-months of every age in the employer declarations; a worker is unlinked when no birth year and sex can be read from the 2023, 2021 or 2019 individual register, which includes workers first registered in Sweden after 2023. Panel B: the tipping point is found by scaling the later-period counts of the band at top-quartile employers by a common factor until τ is zero, verified by a refit at that factor ($\tau = -0.0000$). The allocations add unlinked person-months to the band's cell at the employer and month that declared them, in the interim and later periods as they occur, and leave every other cell as observed; the second allocation adds only the later-period person-months in excess of what each employer's own interim unlinked rate implies, where there is an excess. For the female differential the additions go to young women. The declaration carries no age, so these are counterfactuals on the counts, not a model of who the unlinked workers are, and the tipping point is not a bound. Standard errors clustered by employer.

declarations, before any exclusion by age, linkage or occupation, so it is wider than the employed population the analysis uses. The rate lies between 15.3 and 18.4 per cent in every year from 2020 to 2025 and does not show the rise after 2023 that appears among the young. Most of it is outside the ages the paper studies: workers above 69 hold 13 to 15 per cent of declared person-months, and most of them have no code. Among workers aged 22 to 69 the rate is 3.6 to 6.0 per cent. The rate does rise at the youngest ages once the register ends, from 8.4 per cent at 22 to 25 in 2023 to 12.8 per cent in 2024 and 17.8 per cent in 2025, because workers whose first job begins after the last register was collected have no code at all.

The second is what remains *inside the occupation-linked population used to audit the occupation-classified design*, employed persons aged 22 to 69. That population excludes workers below 22 and above 69 and earners with no occupational record at all. Once the codes of the 2023, 2022 and 2021 registers are carried forward to every year, it is coded almost completely, 99.2 per cent of persons in 2019 rising to 99.9 per cent in each of 2023, 2024 and 2025. Because the rate is conditional on an occupation-linked population, however, it cannot establish that the records missing from that population are harmless.

The third quantity is the one in Panel A of Table A32, where 1.28 per cent is excluded at ages 22–25 in 2019. That is not the same thing: it counts worker-months under each year's own register up to 2022, with codes carried forward from the 2023, 2022 and 2021 registers only from 2023 (the definition of Table A32). Applying that cascade to 2019 as well, the uncoded share in that band and year is 0.87 per cent.

The overall rate is therefore of the order of 15 to 20 per cent in every year and does not rise. What rises is the rate among the youngest workers, and with it the age of the codes of

everyone else, which is why an occupation-classified design deteriorates after 2023 (Part IV) and why the paper’s employment counts use no worker’s own code.

Table A44: Occupation non-match over all employer-declaration person-months, by age, rebuilt from the raw declarations

	2020	2021	2022	2023	2024	2025
<i>Share of person-months with no code under the register rule</i>						
All ages	17.5	16.7	18.4	15.3	16.2	16.6
Aged 22 to 69	5.3	4.8	6.0	3.6	4.2	4.8
Aged 22 to 25	9.0	9.2	9.6	8.4	12.8	17.8
Below 22	41.7	42.6	40.8	38.7	55.5	71.1
Above 69	84.0	80.2	85.6	69.7	65.4	61.1
<i>Share of all person-months</i>						
Aged 22 to 69	81.9	81.7	81.2	79.7	79.6	79.9
Above 69	13.1	13.1	13.0	14.6	14.9	15.4
Declared person-months (millions)	73.9	75.0	78.0	80.7	80.2	39.9

Per cent. The denominator is every distinct employer, person and month in the employer declarations, before any exclusion by age, linkage or occupation. The register rule is that of Table A32, Panel A: each year’s own register up to 2022, and from 2023 the most recent code in the 2023, 2022 and 2021 registers. Age is the calendar year minus the birth year from the 2023, 2021 or 2019 individual register. The 0.2 per cent of person-months with no birth year (Table A42) are in the first row only. 2025 covers the first half of the year. Cells with fewer than five person-months are suppressed before export, 1,043 of 16,715. The paper’s employment counts use no worker’s own occupation code.

Software and model-assisted work. As the paper’s declaration states, the authors used Claude (Anthropic), models of the Opus, Sonnet and Fable families, for code, data processing, editing and analytical review, and ChatGPT (OpenAI) for manuscript review and analytical discussion, reviewed and edited the content as needed, and take full responsibility for it. No model had access to the register data, which never leave Statistics Sweden’s servers.

The posting difference-in-differences is estimated in Python with `pyfixest` (`feols` and `fepois`), and the decile gradient with `linearmodels`. The specifications on the window to December 2025 were independently reproduced in Stata 18.5 with `reghdfe`, matching to three decimal places. The extension to June 2026 was not re-run in Stata. The employment estimates are Poisson pseudo-maximum likelihood fits in R’s `fixest` (`fepois`), called from the Python scripts the replication manifest names, with employer-by-month, employer-by-age and month-by-age effects absorbed.

Access. The administrative micro-level data in the ORU-MICRO-AI database are from several registers of Statistics Sweden, and were obtained after approval from the Swedish Ethical Review Authority (approvals 2021-05040, 2022-03330-02, 2024-01714-0, 2025-04205-02). Researchers and their assistants may access these and similar Swedish micro-level data, subject to approval by SCB. Access is provided through the secure internet-based Micro-data Online Access (MONA) system, and microdata never leave SCB’s servers. Foreign-based researchers may visit a Swedish institution with access or cooperate with researchers in Sweden. Information on the application process is available at <https://>

[//www.scb.se/en/services/ordering-data-and-statistics/ordering-microdata/](https://www.scb.se/en/services/ordering-data-and-statistics/ordering-microdata/) or by contacting SCB at mona@scb.se. Researchers wishing to access our specific data for replication purposes should contact the corresponding author for guidance regarding project approval from the Swedish Ethical Review Authority and SCB.

All replication code for both public and restricted data is available at <https://github.com/Magnus-L/canaries-sweden>. The authors commit to preserving data access and providing reasonable assistance to replication requests for at least five years following publication.

The two parts of the analysis differ for a replicator. Everything on job postings, including the within-employer design of Part V, runs on public advertisements under a CC0 licence and can be reproduced end to end from the repository. The employment results rest on individual-level Swedish register data that never leave Statistics Sweden's servers, so reproducing them requires the approvals listed above. The repository holds the code for both and the exports behind every table and figure, so that anyone without register access can verify what was computed from what.

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